2026 한국응용생명화학회

천연물분과 심포지엄

주제

천연물 연구성과의 산업화 전략

일시 2026. 10. 15.(목) – 16.(금)

장소 한국생명공학연구원 오창분원 국제회의동

주최 한국응용생명화학회 The Korean Society for Applied Biological Chemistry

후원 Waters™ YMC MJ SCIENCE 천연물중앙은행 Natural Product Central Bank NUTRACORE NUTRACEUTICAL FOR BETTER LIFE Jinuchem

2026 한국응용생명화학회

천연물분과 심포지엄

천연물 연구성과의 산업화 전략

2026. 10. 15.(목) – 16.(금)

한국생명공학연구원 오창분원 국제회의동

한국응용생명화학회
The Korean Society for Applied Biological Chemistry

인사말

바쁘신 가운데 참석해 주신 천연물 분야 산·학·연·관 관계자 여러분, 반갑습니다. 한국응용생명화학회 천연물분과 회장을 맡고 있는 이종석입니다.

먼저 천연물분과 심포지엄에 참석해 주신 모든 분들께 진심으로 감사드립니다. 아울러 귀한 강연을 준비해 주신 연사님들과 행사 준비를 위해 애써주신 학회와 분과 운영위원분들께도 깊이 감사드립니다.

지난해에는 충청지회와 공동으로 천연물 소재의 산업화 기술에 대한 심포지엄을 개최했으며, 올해는 천연물분과 단독으로 행사를 마련하게 되었습니다. 특히, 올해는 우리 천연물 연구자들이 함께 고민해야 할 중요한 질문을 중심으로 프로그램을 구성했습니다.

바로 우리가 연구해 온 천연물 연구성과는 얼마나 산업으로 연결되고 있는가? 라는 질문입니다.
우리나라에는 우수한 천연물 연구자와 축적된 연구성과, 그리고 다양한 소재와 기술이 있습니다. 그럼에도 불구하고 연구성과가 실제 기업의 제품과 매출, 새로운 산업적 가치로 이어지는 사례는 아직 충분하지 않다는 아쉬움이 있습니다.
연구성과가 산업으로 연결되기까지는 특허와 지식재산권, 원료의 표준화와 품질관리, 인허가, 기술이전, 생산공정과 사업성 등 넘어야 할 과제가 많기 때문입니다.

그래서 저는 여러분께 한 가지 화두를 던지고 싶습니다.
천연물 연구의 산업화는 연구가 끝난 다음에 고민해야 하는 것일까요?
저는 연구를 시작하는 단계에서부터 산업화를 함께 생각할 필요가 있다고 봅니다.
어떤 소재를 연구할 것인지 결정하는 단계에서부터 이 소재가 어느 산업에서 활용될 수 있는지, 기업이 필요로 하는 데이터는 무엇인지, 그리고 연구성과를 실제 제품과 기술로 발전시키기 위해 무엇을 준비해야 하는지 고민하자는 것입니다.

물론 모든 연구가 처음부터 산업화를 목표로 해야 한다는 의미는 아닙니다. 기초연구와 새로운 과학적 발견 역시 매우 중요합니다. 다만 연구 초기부터 산업적 활용 가능성을 함께 살펴본다면, 우리가 만들어내는 연구성과의 가치와 활용 범위를 더욱 넓힐 수 있을 것입니다.

이러한 문제의식을 담아 올해 심포지엄의 주제를 「천연물 연구성과의 산업화 전략」으로 정했습니다.
오늘 심포지엄은 크게 기술강연과 초청강연으로 구성했습니다.
기술강연에서는 최신 연구동향과 천연물 분석기술, LC-HRMS 분석 워크플로 등 연구현장에서 실제로 활용할 수 있는 기술적 내용을 준비하였습니다. 초청강연에서는 특허와 IP 확보 전략, 용도별 실험 데이터, 식품산업화와 시장진입 전략, 기술이전 및 사업화 사례, 그리고 Scale-up 전략까지 연구성과의 산업화 과정 전반을 각 분야 최고 전문가들과 함께 살펴보고자 합니다.

이 자리가 단순히 새로운 연구결과를 공유하는 데 그치지 않고, 나의 연구가 산업에서 활용되려면 무엇이 더 필요할까? 라는 질문을 각자의 연구에 적용해 보는 기회가 되기를 바랍니다.
또한 연구자와 기업, 기술이전 전문가 등 다양한 분야의 관계자들이 서로의 관점과 수요를 이해하고 새로운 협력의 기회를 만드는 자리가 되었으면 합니다.

천연물 연구의 미래는 우수한 연구성과를 만들어내는 데서 한 걸음 더 나아가, 그 성과를 산업과 사회의 가치로 연결할 때 더욱 큰 가능성을 열 수 있다고 생각합니다.
끝으로 참석해 주신 모든 분들께 다시 한번 감사드리며, 오늘 심포지엄이 우리 천연물 연구의 새로운 도약과 산업화의 가능성을 여는 뜻깊은 자리가 되기를 기원합니다.

감사합니다.

2026년 10월

한국응용생명화학회 천연물분과 회장 이 종 석

목차

인사말 2

행사 개요 및 안내 5

프로그램 6

구두 발표 8

구두 발표 목록 9

구두 발표 초록 10

포스터 발표 22

포스터 발표 목록 23

포스터 초록 28

행사 개요 및 안내

행사명	2026 한국응용생명화학회 천연물분과 심포지엄
주제	천연물 연구성과의 산업화 전략
일시	2026년 10월 15일(목) – 16일(금)
장소	한국생명공학연구원 오창분원 국제회의동
주최	한국응용생명화학회 천연물분과

행사장 안내

주소	충북 청주시 청원구 오창읍 연구단지로 30, 한국생명공학연구원 오창분원 국제회의동
구두발표	국제회의동
포스터	로비 (10.15 14:30–15:30 / 10.16 09:00–10:00)
교통 안내	한국생명공학연구원 홈페이지 오시는 길 https://portal.kribb.re.kr/board?menuId=MENU00923&siteId=null

프로그램

DAY 1 2026. 10. 15.(목)

기술강연	09:30–10:10	Waters 솔루션을 활용한 최신 연구 동향	**김현우 이사** Waters
	10:10–11:00	복잡한 천연물 시료의 성분 해석을 위한 LC-HRMS 분석 워크플로와 산업적 활용 사례	**최사무엘 부장** Waters
	11:00–11:50	천연물 분석에서 재현성과 데이터 품질을 향상시키는 LC 컬럼 및 바이알 솔루션	**박미진 부장** Waters
점심	12:00–13:15	점심 (미제공)	
개회	13:15–13:30	개회 및 축사, 단체 사진	
초청강연 I 연구성과의 IP 확보	13:30–14:00	천연물 연구의 최신 특허 전략	**배지현 변리사** 특허법인(유) 화우
	14:00–14:30	천연물 특허 확보를 위한 용도별 실험 데이터 (화장품, 식품 중심)	**지예은 변리사** 큰담특허법률사무소
포스터	14:30–15:30	포스터 전시	로비
초청강연 II 연구성과의 산업화	15:30–16:00	천연물 소재의 식품 산업화: 제도 활용과 시장진입 전략	**이미영 주무관** 식품의약품안전처
	16:00–16:30	천연물 분야 기술 사업화(기술이전) 전략 및 사례	**최범락 대표** 뉴트라코어
	16:30–17:00	천연물 소재의 산업화를 위한 Scale-up 전략: 산삼 배양근 대량생산 사례	**박소영 교수** 충북대학교
네트워킹	17:00–17:30	천연물 분과 총회	
	18:00–20:00	만찬, 네트워킹	TBA

※ 장소 표기가 없는 프로그램은 국제회의동에서 진행됩니다. 세부 프로그램은 변경될 수 있습니다.

DAY 2 2026. 10. 16.(금)

포스터	09:00–10:00	포스터 전시 로비
학생 발표	10:00–11:00	학생 구두발표
폐회	11:00–11:30	시상 및 폐회

발표 안내

구두발표	기술강연 3편 · 초청강연 5편 · 학생 구두발표 4편	국제회의동
포스터 발표	총 32편	로비 전시
시상 및 폐회	10.16(금) 11:00–11:30	국제회의동

※ 세부 프로그램은 변경될 수 있습니다.

구두 발표

기술강연	10.15(목) 09:30–11:50
초청강연 I	10.15(목) 13:30–14:30
초청강연 II	10.15(목) 15:30–17:00
학생 구두발표	10.16(금) 10:00–11:00

구두 발표

기술강연

Emerging Trends in Metabolomics and Spatial Molecular Analysis Using Advanced Mass Spectrometry
10.15(목) 09:30–10:10
김현우 이사 Waters

복잡한 천연물 시료의 성분 해석을 위한 LC-HRMS 분석 워크플로와 산업적 활용 사례
10.15(목) 10:10–11:00
최사무엘 부장 Waters

MaxPeak Premier Microflow 컬럼으로 높이는 LC-MS 감도와 지속가능성
10.15(목) 11:00–11:50
박미진 부장 Waters

초청강연 I · 연구성과의 IP 확보

천연물 연구의 최신 특허 전략
10.15(목) 13:30–14:00
배지현 변리사 특허법인(유) 화우

천연물 특허 확보를 위한 용도별 실험 데이터 (화장품, 식품 중심)
10.15(목) 14:00–14:30
지예은 변리사 큰담특허법률사무소

초청강연 II · 연구성과의 산업화

Regulatory Trends and Safety Assessment for Innovative Food Ingredients: From Cell-Based Foods and Precision Fermentation
10.15(목) 15:30–16:00
이미영 주무관 식품의약품안전처

Individual Recognition and Commercialization Strategy for Health Functional Food Ingredients: A Case Study of Wasabi Leaf Extract
10.15(목) 16:00–16:30
최범락 대표 ㈜뉴트라코어

천연물 소재의 산업화를 위한 Scale-up 전략: 산삼 배양근 대량생산 사례
10.15(목) 16:30–17:00
박소영 교수 충북대학교

학생 구두발표

Khellactone-Type Coumarins from *Peucedanum japonicum*: Isolation, Neuroinflammation-Related Screening, and Network Pharmacology
10.16(금) 10:00–11:00
박정현 경북대학교

Comparative Metabolomic Analysis of *Vitex rotundifolia* L. f. Across Plant Parts and Geographic Locations in Jeju Island and Its Association with Environmental Characteristics
10.16(금) 10:00–11:00
유슬범 한국생명공학연구원

Comparative Micromorphology of Adhesive and Aerial Roots and Metabolite Profiles across Species and Organs in Climbing Plants
10.16(금) 10:00–11:00
백신유 국립생태원

Lancemaside A isolated from *Codonopsis lanceolata* ameliorates inflammation-induced intestinal barrier dysfunction
10.16(금) 10:00–11:00
정현정 강원대학교

김현우 이사

Waters Korea ANR (Academic & Research) Team Leader

주요 경력

Waters Korea Academic & Research 팀장
질량분석 및 분석과학 분야의 연구·응용 및 산학협력 활동

전문 분야

고분해능 질량분석(HRMS), 대사체학(Metabolomics), 이온 모빌리티(Ion Mobility), 질량분석 이미징(MSI) 및 천연물 분석

학회 활동

질량분석이미징연구회 산학이사 및 창립 핵심위원
아시아 질량분석이미징(MSI) 커뮤니티 핵심 멤버
대한화학회 산학협력 이사 (2026–현재)
한국단백체학회(KHUPO) 산학협력 이사 (2023–현재)

주요 수상

한국분석과학회 분석기술상 (2025)
Waters Asia Sales Award (2024)

Emerging Trends in Metabolomics and Spatial Molecular Analysis Using Advanced Mass Spectrometry

Hyunwoo Kim*

Waters Korea, Gwangmyeong, Republic of Korea

Recent advances in mass spectrometry have expanded its applications in metabolomics and spatial molecular analysis. High-resolution mass spectrometry (HRMS) enables accurate mass measurement and molecular identification, while ion mobility provides additional separation and structural information for complex molecules. In addition, desorption electrospray ionization mass spectrometry imaging (DESI-MSI) enables direct visualization of the spatial distribution of metabolites and lipids in biological samples with minimal sample preparation. This presentation introduces recent analytical approaches using HRMS, ion mobility, and DESI-MSI, with applications in metabolomics, natural product research, and spatial molecular analysis.

Keywords Metabolomics, High-resolution mass spectrometry, Ion mobility, DESI-MSI, Spatial molecular analysis

최사무엘 부장

Waters Korea HRMS Market Development Team Leader

학력

University of Maryland, College Park 화학 학사 (2010)
George Mason University 화학 석사 (2012)
University of Maryland, College Park 화학/분석화학 박사 (2021)

경력

Waters Korea HRMS Market Development Team Leader (2025–현재)
Shimadzu MS Technical Support Scientist (2024–2025)
AB Sciex Biologics Workflow Specialist (2023–2024)
AstraZeneca Senior Scientist (2022–2023)
Bruker Daltonics Field Application Scientist (2021–2022)

전문 분야

LC-MS, 고분해능 질량분석(HRMS), 단백질 특성분석, 프로테오믹스, 펩타이드체학, 대사체학, CE-MS, 바이오의약품 분석

주요 수상

Glen E. Gordon Memorial Award (2020)
US HUPO Student Travel Award (2017)

복잡한 천연물 시료의 성분 해석을 위한 LC-HRMS 분석 워크플로와 산업적 활용 사례

Sam Choi[1*]

[1]Waters Korea, Gwangmyeong, Republic of Korea

천연물 시료는 다양한 화합물이 복잡한 매트릭스 내에 존재하기 때문에, 주요 성분의 확인뿐만 아니라 미지 성분의 탐색 및 구조적 특성 규명을 위해 종합적인 분석 전략이 요구됩니다. 특히 최근 천연물 연구는 단순한 지표성분 분석을 넘어 전체 성분 프로파일링, 기능성 물질 발굴 및 원료의 품질관리로 확대되고 있으며, 이에 따라 액체크로마토그래피-고분해능 질량분석법(LC-HRMS)을 활용한 비표적 분석의 중요성이 더욱 증가하고 있습니다.

본 발표에서는 복잡한 천연물 시료의 성분 해석을 위한 LC-HRMS 기반 분석 워크플로를 소개합니다. 시료 전처리 및 액체크로마토그래피 분리부터 고분해능 MS 및 MS/MS 데이터 획득, 정확질량을 이용한 분자식 추론, MS/MS fragmentation 정보를 기반으로 한 구조 정보 확보, 데이터베이스 및 스펙트럼 라이브러리를 활용한 화합물 동정에 이르는 전반적인 분석 과정을 다룹니다. 또한 복잡한 시료에서 다수의 화합물 정보를 효율적으로 추출하고 시료 간 성분 차이를 비교하기 위한 데이터 처리 및 해석 전략을 소개합니다.

아울러 천연물 원료 및 제품의 성분 프로파일링, 품질관리, 기능성 소재 발굴 및 표준화 등 산업 현장에서 LC-HRMS가 활용되는 사례를 소개합니다. 다양한 천연물 시료의 분석 사례를 바탕으로 복잡한 시료에서 확보한 화학적 정보를 종합적으로 해석하고, 이를 유효성분 및 품질지표 탐색과 같은 산업적 활용으로 연결하는 분석 접근법을 제시합니다. 이를 통해 LC-HRMS 기반 분석이 천연물 연구의 탐색적 분석을 넘어 소재 개발 및 품질관리 등 실제 산업적 응용 분야에서도 활용될 수 있는 가능성을 제시합니다.

Keywords 천연물, LC-HRMS, 비표적 분석, 성분 프로파일링, 고분해능 질량분석

박미진 부장

워터스코리아 컬럼 및 소모품 마케팅 부장

학력

가천대학교(글로벌캠퍼스) 화학공학 학사 (2006)
가천대학교(글로벌캠퍼스) 화학공학 석사 (2008)

경력

워터스코리아 컬럼 및 소모품 마케팅 부장 (2008–현재)

주요 강의

Scaling Up Analytical Method from UPLC to HPLC using MaxPeak Premier Columns (KDRA 한국의약분석연구회, 2024)
GLP-1을 비롯한 바이오의약 분석을 위한 최신 크로마토그래피 컬럼 소개 (2025 한국의약분석연구회 워크숍, 2025)

MaxPeak Premier Microflow 컬럼으로 높이는 LC-MS 감도와 지속가능성

박미진

Waters Korea

LC-MS 기반 정량 바이오분석, 신약 개발, 오믹스 연구에서는 감도를 더 높이는 동시에 귀한 시료와 용매를 아끼는 방법이 필요합니다. 마이크로플로우(1.0 mm ID, 0.3mm ID) LC는 이런 요구에 맞는 대안이지만, 기존 스테인리스 스틸 하드웨어에서는 금속 표면과의 비특이적 흡착 때문에 회수율과 재현성을 확보하기 어려웠습니다.

MaxPeak Premier Microflow 컬럼은 MaxPeak HPS(High Performance Surfaces) 기술을 적용해 MaxPeak Premier 2.1 mm ID 컬럼 수준의 감도와 회수율을 구현하도록 설계되었습니다. 이 컬럼은 스테인리스 스틸 마이크로플로우 컬럼보다 최대 2배 높은 감도를 제공하며, 2.1 mm ID 컬럼과 비교해 시료 사용량은 최대 75%, 용매 사용량은 최대 4배까지 줄일 수 있습니다. 또한 프로테오믹스에서 다루기 까다로운 산성 펩타이드와 인산화 펩타이드도 첫 번째 주입부터 높은 회수율과 대칭적인 피크 형태를 보여, 컬럼 컨디셔닝에 드는 시간을 줄일 수 있었습니다

이 세미나에서는 이러한 사례를 바탕으로 기존 2.1 mm 분석법을 MicroFlow column으로 옮길 때 고려할 점을 살펴보고, 감도, 시료 절약, 지속가능성을 함께 개선할 수 있는 실용적인 방법을 공유합니다.

배지현 변리사

특허법인(유) 화우 파트너 변리사

학력

이화여자대학교 자연과학대학 생물과학과 졸업 (2003)
연세대학교 공과대학 생명공학과 석사 (2009)

경력

식품의약품안전청 생물의약품평가부 (2004)
제41회 변리사 시험 합격 (2004)
5T국제특허법률사무소, 박장원특허법률사무소 변리사
특허청 한국지식재산전략원 전문위원 (2011–2012)
삼성서울병원 R&D 사업화추진본부 책임연구원 (2013–2015)
특허법인(유) 화우 파트너 변리사 (2016–현재)

주요 활동

지식재산처 국가 R&D 특허동향조사사업, 지재권중심기술획득전략사업 등 정부과제 수행책임자
영국 Oxford 대학 Isis Innovation Technology Commercialization Program 수료
대한변리사회 정회원, 기술거래사
특허심판원 지재심판연구회 회원, 서울지방법원 전문심리위원

주요 논문

「진단방법발명에 관한 특허성 연구」 (연세대학교, 2009)

천연물 연구의 최신 특허 전략

배지현

특허법인(유) 화우

천연물 기반 특허는 추출물, 균주 등 다양한 기술 카테고리들을 커버하고 있으며, 기술의 속성상구조를 직접 특정하기 어려우며, 전통지식에 기반한 선행기술이 광범위하여, 합성신약과는 본질적으로 다른 특허 전략을 요구한다.

최근 몇년간 미국, 유럽, 중국, 일본 등 주요국에서 이슈된 심판, 분쟁 사례를 살펴보면 자연산물의 특허적격성, 명확성 등의 기재요건 불비, 전통의학 이론에 기초한 진보성 등이 특허등록을 위해 주요하게 논의된 요건들이다. 이러한 케이스들의 판단 논리들을 살펴보고, 발명의 구조, 물성, 제법과 같은 주요 카테고리의 특허 청구항 설계와 특허 포트폴리오 전략 설계원리를 제시한다.

주제어 천연물 의약품, 건강기능식품, 특허 적격성, 제조방법 한정 청구항

지예은 변리사

큰담특허법률사무소 대표변리사

학력

서울대학교 생물교육과 학사 (2001)
건국대학교 기술경영학과 박사 (2026)

경력

법무법인 KCL 변리사 (2009–2011)
새온특허법률사무소 변리사 (2011–2014)
한국발명진흥회 전문위원 (2015–2016)
화인특허법률사무소, 특허법인 서한 대표변리사 (2016–2023)
큰담특허법률사무소 대표변리사 (2024–현재)

주요 기술분야

바이오(화장품, 식품), 제약, 화학, 화공, 소재, 의료기기

자격 및 활동

변리사 시험 합격 (2008), 기술거래사 (2016)
지식재산처
산업재산권분쟁조정위원회 조정위원 (2016–현재)
중소벤처기업부
중소기업기술분쟁조정중재위원회 조정위원 (2018–현재)
해양수산부 해양수산과학기술진흥원 비상임이사 (2023–현재)
기후부 국립생물자원관 고문변리사 (2024–현재)
대한변리사회 공보이사 (2018–2019), 대의원 (2018–2021)

저서 및 연구 보고서

쟁점별 직무발명 최신 판례 2026 (지식재산처, 한국발명진흥회, 2026)
직무발명 보상금 판결 분석 (특허청, 2024)
산업별 상표출원 동향 분석: 음료제조업/카페 (특허청, 한국지식재산연구원, 2024)

천연물 특허 확보를 위한 용도별 실험 데이터 (화장품, 식품 중심)

지예은

큰담특허법률사무소

발표 내용 요약

1. 천연물 연구 결과의 특허 확보 전략 (추출물/small molecule)
2. 화장품 특허에서 필요한 실험 데이터
3. 식품 특허에서 필요한 실험 데이터
4. 그 외 특허(의약품, AI 분석)에서 필요한 실험 데이터

이미영 주무관

식품의약품안전처 식품기준과

학력

Master's degree in food engineering, Chungbuk National University

Doctor's degree in food engineering, Chungbuk National University

Master's degree, Chungnam National University Patent Law School

자격

Food technician

Fishery product manufacturing technician

Distribution manager

Agricultural product quality manager, etc.

경력

Scientific Officer, Food Standard Division, Ministry of Food and Drug Safety (현재)

Researcher, National Institute of Crop Science, National Institute of Horticultural and Herbal Science

Professional Researcher, Functional Food Department, National Institute of Agricultural Sciences

Instructor of Food Engineering, Chungbuk Provincial University, Kyungpook National University

Ministry of Food and Drug Safety Dietary and Nutritional Safety Policy and Policy Research Institute

Health Researcher, Nutritional Function Research Department, Korea Food and Drug Safety Evaluation Institute

Adjunct Professor, Korea University Sejong Campus

Regulatory Trends and Safety Assessment for Innovative Food Ingredients: From Cell-Based Foods and Precision Fermentation

Mi-Young Lee*

Food Standard Division, Ministry of Food and Drug Safety, Republic of Korea

Advances in cell culture and precision fermentation are expanding opportunities to develop food ingredients, while research on natural and botanical resources continues to support innovation foods. Translating these developments into marketable products requires early integration of regulatory requirements, ingredient characterization, and scientific evidence of safety.

This research reviews international regulatory trends for emerging food technologies and outlines relevant pathways administered by the Ministry of Food and Drug Safety (MFDS) in the Republic of Korea. Particular attention is given to the temporary recognition of standards and specifications for new food ingredients and the individual recognition of functional ingredients for health functional foods. The regulatory process is discussed from ingredient classification and dossier preparation to scientific review, responses to requests for additional information, and recognition. Key development considerations include source identification, history of food use, manufacturing controls, compositional characterization, specifications, validated analytical methods, stability, and batch consistency.

For botanical ingredients, species authentication, plant parts used, extraction conditions, and standardization of functional or marker constituents are emphasized. For cell-based foods, assessment focuses on cell sources and characteristics, culture media, scaffolds where used, and potential process-related residues. For precision fermentation-derived ingredients, production strain characterization, genetic modifications where applicable, purification, and residual microbial components are addressed. Safety assessment approaches encompass hazard identification, dietary exposure estimation, nutritional assessment, allergenicity evaluation, and toxicological testing appropriate to the ingredient and applicable requirements. For health functional food development, additional emphasis is placed on substantiating the proposed physiological benefits through well-designed human intervention studies, supported by mechanistic evidence, and establishing appropriate daily intake levels and conditions of use. Integrating these considerations throughout research and development can improve the quality of regulatory submissions and support the development of safe, scientifically substantiated food ingredients.

Keywords cell-based foods, precision fermentation, food ingredient regulation, safety assessment

최범락 대표

(주)뉴트라코어 대표이사

학력

단국대학교 미생물학과 이학사 (1999)
단국대학교 미생물학과 이학석사 (2001)
성균관대학교 식품생명공학 이학박사 (2021)

주요 이력

(주)뉴트라코어 대표이사 (2018–현재)
농기평 심의위원 (2022–현재)
건강기능식품협회 기술분과위원 (2015–2018)
(주)에이치엘사이언스 연구소장 (2012–2018)
서울대학교 지능형생체계면공학연구센터 연구원 (2002–2003)

전문 분야

천연물 기능성 원료 개발, 건강기능식품 개별인정, 원료 표준화, 인체적용시험, 식약처 인허가, GMP 생산 연계

주요 실적

건강기능식품 기능성 원료 개발 및 식약처 개별인정 업무 20년 이상 수행 — 헛개과병추출물, 석류농축액, 옥수수수염·레몬밤추출복합물, 와사비잎추출물 등 다수의 개별인정 원료 개발
천연물 소재의 지표성분 설정, 기준규격 확립, 비임상시험, 인체적용시험, 식약처 인허가 및 사업화 전략 수립
와사비잎추출물, 옥수수수염·레몬밤추출복합물, 석류, 레몬밤, 민들레 등 천연물 기능성 소재 관련 논문 약 30편
(주)뉴트라코어에서 연구개발부터 원료 표준화, 효능평가, CRO, 식약처 인허가, GMP 생산까지 전 과정 총괄

Individual Recognition and Commercialization Strategy for Health Functional Food Ingredients: A Case Study of Wasabi Leaf Extract

Beom Rak Choi

Nutracore Co., Ltd., Suwon 16514, Republic of Korea

The commercialization of natural materials requires an integrated development strategy that goes beyond efficacy screening. It must include verification of raw material origin, selection of marker compounds, standardization of manufacturing processes, safety evaluation, preclinical and human studies, individual recognition by the Ministry of Food and Drug Safety (MFDS), and product development strategy. Nutracore has established a one-stop development platform that connects natural material research, raw material standardization, analytical method development and validation, clinical research organization (CRO) services, regulatory consulting, and GMP manufacturing to support the commercialization of health functional food ingredients.

In this presentation, we introduce the development and commercialization strategy of a natural health functional food ingredient, focusing on wasabi leaf extract. Wasabi leaf extract is prepared from the leaves of Wasabia japonica (Miq.) Matsum., and isovitexin was selected as the marker compound. During development, the food-use history and domestic/international use information of the raw material were reviewed, and the extraction and manufacturing processes were standardized based on marker compound content and manufacturing yield. In addition, product specifications, HPLC-based analytical method validation, and safety-related data were established to support the MFDS individual recognition application.

To evaluate functionality, a preclinical study was conducted using a high-fat diet-induced obese ICR mouse model. Body weight, body fat mass, abdominal fat mass, blood lipid levels, and markers related to lipid synthesis and metabolism were assessed. Subsequently, a human study was conducted in Korean adults with a BMI of 25–32 kg/m² to evaluate the effects of 250 mg/day of wasabi leaf extract for 12 weeks. The study showed improvements in major body composition parameters, including body fat mass, body fat percentage, body weight, and BMI.

Based on these research data, wasabi leaf extract was developed as an individually recognized health functional food ingredient for body fat reduction and was further commercialized through finished product development and distribution channel application. In addition, other Nutracore-developed individually recognized health functional food ingredients, such as corn silk and lemon balm extract complex, have also been applied to commercial products and reported in official production performance data.

Keywords natural materials, individually recognized health functional food ingredient, wasabi leaf extract, raw material standardization, human study, commercialization

박소영 교수

충북대학교 농업생명환경대학 원예과학과 교수

학력

부산대학교 이학사 (1993)
충북대학교 원예학 농학석사 (1997)
충북대학교 원예학 농학박사 (2002)

경력

충북농업기술원 농업연구사 (1993–1998)
국립산림과학원 임업연구사 (2003–2012)
University of British Columbia (캐나다) Visiting Scientist (2007–2009)
충북대학교 원예과학과 교수 (2012–현재)

주요 활동

한국화훼학회 부회장 (2021–현재)
한국식물생명공학회 부회장 (2022–2024)
Plant Cell, Tissue and Organ Culture 편집위원 (2020–현재)
국제원예학회(ISHS) Working Group Chair (2018–2022)

천연물 소재의 산업화를 위한 Scale-up 전략: 산삼 배양근 대량생산 사례

박소영[1*], 이한솔[1], 이인영[1], H.N. Murthy[1,2], 이효신[1], 권수정[1], 권아름[3], 백기엽[3]

[1]충북대학교 농업생명환경대학 원예과학과, 청주, 28644, 대한민국
[2]CHRIST University, 인도, [3](주)웰그린, 대한민국

식물 생명공학과 천연물 연구의 융합은 약용작물을 고부가가치 기능성 소재 생산을 위한 지속 가능한 바이오 팩토리로 활용할 수 있는 혁신적인 기회를 제공한다. 특히 식물 세포, 조직 및 기관 배양 플랫폼은 기존 농업 재배 방식의 원료 수급 불확실성과 환경적 제약을 극복하는 동시에, 식품 및 의약품 산업에서 요구하는 원료의 균일성과 안전성을 효과적으로 확보할 수 있게 한다. 본 연구에서는 희귀한 자연산 산삼(*Panax ginseng*)으로부터 세포 분화 유도 기법을 활용하여 안정적으로 관리 가능한 세포주 및 부정근 라인을 구축하였다. 아울러 이차대사산물의 생합성 및 축적을 극대화하기 위해 유도처리(elicitation) 기법이 적용된 생물반응기 시스템을 최적화하였다. 대사체 프로파일링 분석 결과, 구축된 산삼 배양근에서는 일반 재배 인삼에서는 드물게 검출되는 다양한 미량 진세노사이드가 축적되었으며, 이를 바탕으로 신규 생리활성 화합물의 분리 및 구조적 특이성을 규명하였다. 나아가 10톤 규모의 산업용 파일럿 생물반응기를 통한 Scale-up 공정을 성공적으로 달성하여 8주 배양 주기에 1000 kg/ton의 수율을 확보하였다. 특히 150년근 산삼 유래 배양근으로부터 세포 재프로그래밍을 통해 체세포배를 성공적으로 유도함으로써, 희귀 유전자원의 장기 보존과 대량 증식을 위한 견고한 기틀을 마련하였다. 종합하여 볼 때, 본 연구에서 제시한 통합적 접근법은 산삼 배양근 연구의 지속 가능성과 안전성을 높이고, 다양한 산업적 응용을 실현하기 위한 미래지향적 패러다임을 제시한다.

Keywords 산삼, 배양근, 진세노사이드, 생물반응기

Acknowledgement 본 연구는 농촌진흥청 식물세포기반 고부가 기능성소재 대량생산 원천기술 개발 사업(RS-2026-25512707)의 지원에 의해 이루어진것임.

Khellactone-Type Coumarins from *Peucedanum japonicum*: Isolation, Neuroinflammation-Related Screening, and Network Pharmacology

Jeong Hyun Park[1], Woo-Cheol Shin[1], Jin-Kyu Jang[1], Hyeon Seon Na[1], Yong Jae Jeong[1], Dong-Sung Lee[2] and Dae Young Lee[1*]

[1]BK21 FOUR KNU Creative BioResearch Group, School of Life Sciences, Kyungpook National University, Daegu 41566, Republic of Korea
[2]Research Institute of Pharmaceutical Sciences, College of Pharmacy, Chosun University, Gwangju 61452, Republic of Korea

Phytochemical investigation of *Peucedanum japonicum* aerial parts was conducted to characterize khellactone constituents and evaluate candidate molecular targets for neuroinflammation. Chromatographic separation of the ethyl acetate fraction led to the identification of eighteen khellactone-type dihydropyranocoumarins (1–18). Their chemical structures and absolute configurations were elucidated using 1D/2D NMR, high-resolution UPLC-QTOF-MS^E fragmentation, and electronic circular dichroism (ECD) analysis, confirming a common (3′*S*,4′*S*) configuration. Network pharmacology identified 72 predicted targets common to the compounds, 50 of which overlapped with neuroinflammation-associated targets. Protein–protein interaction analysis identified six hub genes: STAT3, BCL2, HIF1A, GSK3B, PPARA, and MMP9. Molecular docking predicted a favorable interaction between compound 13 and PPARA. In BV2 microglial cells, several isolates inhibited lipopolysaccharide-induced nitric oxide production, with IC_{50} values of 5.5–6.8 μM. No clear protective effects were observed in HT22 cells.

These findings characterize the khellactone constituents of *P. japonicum* and identify compounds with potential anti-neuroinflammatory activity. The predicted molecular targets require experimental validation.

Keywords *Peucedanum japonicum,* Khellactones, Network pharmacology, Molecular docking, Anti-neuroinflammatory

Comparative Metabolomic Analysis of *Vitex rotundifolia* L. f. Across Plant Parts and Geographic Locations in Jeju Island and Its Association with Environmental Characteristics

Seul Beom Yu[1,2], Hyoung-Geun Kim[1], Hangyeol Lee[1], Doo-Young Kim[1], In Soo Suh[3], Sung Chun Kim[3], Sei-Ryang Oh[1], Dongho Lee[2*], Hyung Won Ryu[1*]

[1]Natural Product Research Center, Korea Research Institute of Bioscience and Biotechnology KRIBB, 30-Yeongudanji-ro, Ochang-eup, Cheongwon-gu, Cheongju-si, Chungbuk, 28116, Republic of Korea
[2]Department of Plant Biotechnology, College of Life Sciences and Biotechnology, Korea University, 145-Anam-ro, Seongbuk-gu, Seoul, 02841, Republic of Korea
[3]Biodiversity Research Institute, Clean Bio Business Division, JejuTechnopark, 338, Sinryedong-ro, Namwon-eup, Seogwipo-si, Jeju-do, 63608, Republic of Korea

Vitex rotundifolia L. f. is a coastal halophyte belonging to the Verbenaceae family and is widely distributed across coastal regions of Korea, including Jeju Island, as well as in China, Japan, Southeast Asia, and Australia. Its seeds have long been used for medicinal purposes, and the plant is known to contain various secondary metabolites, including iridoids, flavonoids, phenolic compounds, and diterpenoids. The composition and accumulation of these secondary metabolites can vary depending on biotic and abiotic factors. Therefore, this study compared the metabolite profiles of *V. rotundifolia* across different geographic locations and plant parts on Jeju Island to identify useful metabolites for raw material standardization and to elucidate their relationships with environmental characteristics. Samples were collected from ten locations on Jeju Island, and the metabolite profiles of leaves, stems, immature seeds, and mature seeds were analyzed using UPLC-QTOF/MS. Multivariate statistical analysis was performed to evaluate differences in metabolite composition among plant parts and to identify key compounds contributing to these variations. Based on the analytical results and their potential utility, two major iridoid compounds were selected as key chemical markers for raw material standardization. Comparison of their contents across collection sites revealed regional differences in the relative abundance of these two compounds. These findings provide fundamental data for evaluating the relationships between biotic and abiotic factors and key metabolites, and for establishing raw material standardization criteria to support the industrial utilization of *V. rotundifolia*.

Keywords *Vitex rotundifolia* L. f., Iridoids, Metabolomics, Geographic variation, Soil characteristics

Comparative Micromorphology of Adhesive and Aerial Roots and Metabolite Profiles across Species and Organs in Climbing Plants

Shin-yu Baek[1], Haejin Bae[1*]

[1]*Ecological Technology Research Team, Division of Restoration Ecology Research, National Institute of Ecology, Seocheon 33657 Republic of Korea*

Some climbing plants have adhesive roots or aerial roots, specialized organs responsible for attachment. Because of this adhesive function, the attachment mechanisms and adhesive substances of these organs have attracted considerable attention in the field of biomimicry. Although adhesive roots are found in a range of climbing plant species, previous studies in this area have focused predominantly on the adhesive roots of *Hedera helix*. Therefore, this study aimed to examine interspecific differences in the morphology of adhesive and aerial roots and to compare metabolite profiles across species and organs. The micromorphological characteristics of these roots and metabolite profiles across species and organs were compared in *H. helix*, *Trachelospermum asiaticum*, and *Epipremnum aureum* using high-resolution stereomicroscopy, LV-SEM, and LC-MS/MS. The results showed that although the adhesive roots of *H. helix* and *T. asiaticum* differed in morphology, irregular hair-like microstructures were observed in both species. In *E. aureum*, elongating aerial roots showed a relatively regular microstructure. Metabolite profiling revealed differences across species and organs, with differences among species more evident than those among organs. These findings suggest interspecific variation in the micromorphology of adhesive and aerial roots, as well as variation in metabolite profiles across species and organs. They may contribute to a broader understanding of attachment traits in climbing plants and provide a basis for the future development of bio-derived adhesive materials and biomimetic technologies inspired by plant attachment traits.

Keywords climbing plant, adhesive root, micromorphology, metabolomics, biomimicry

Lancemaside A isolated from *Codonopsis lanceolata* ameliorates inflammation-induced intestinal barrier dysfunction

Hyeon-Jeong Jeong[1], So-Ri Son[2], Dae Sik Jang[2] and Hyun Bong Park[1,*]

[1]*Department of Biological Sciences, Kangwon National University, Gangneung 25457, Republic of Korea*
[2]*Department of Biomedical and Pharmaceutical Sciences, Graduate School, Kyung Hee University, Seoul 02447, Republic of Korea*

Codonopsis lanceolata is an edible plant traditionally consumed as both a food and herbal medicine, yet the biological activities of its aerial parts remain largely unexplored. Compounds isolated from the aerial parts were screened for anti-inflammatory activity, with Lancemaside A showing the strongest activity among the tested compounds. Lancemaside A reduced inflammatory responses in intestinal epithelial cells and improved barrier integrity, as shown by increased transepithelial electrical resistance (TEER), decreased paracellular permeability, and recovery of tight junction protein expression. Transcriptomic analysis showed changes in pathways related to epithelial barrier function and antioxidant responses, along with increased expression of tight junction- and Nrf2-associated genes. ARE-luciferase and protein expression analyses further showed that Lancemaside A activates Nrf2-associated signaling. These results suggest that Lancemaside A is an active constituent of *C. lanceolata* aerial parts with intestinal barrier-protective properties and potential as a food-derived functional ingredient for intestinal health.

Keywords *Codonopsis lanceolata*, lancemaside A, inflammation, intestinal barrier, tight junction, Nrf2

포스터 발표

Day 1	10.15(목) 14:30–15:30
Day 2	10.16(금) 09:00–10:00

포스터 발표

P-01 **Lancemaside A isolated from *Codonopsis lanceolata* ameliorates inflammation-induced intestinal barrier dysfunction**

Hyeon-Jeong Jeong[1], So-Ri Son[2], Dae Sik Jang[2] and Hyun Bong Park[1,*]

[1]*Department of Biological Sciences, Kangwon National University, Gangneung 25457, Republic of Korea*
[2]*Department of Biomedical and Pharmaceutical Sciences, Graduate School, Kyung Hee University, Seoul 02447, Republic of Korea*

P-02 **Standardization and Biological Activity Screening of *Cakile edentula* (Bigelow) Hook. Leaf Extract**

Jun Seok Wang[1], Chul Woong Ho[1], Hyeon-Jeong Jeong[1], Ye Jin Kim[1], Aram Kim[1], Kyung-Won Min[1], and Hyun Bong Park[1*]

[1]*Department of Biological Sciences, College of Natural Sciences, Kangwon National University, Gangneung, 25457, Republic of Korea*

P-03 **Bioactive potential of *Codium fragile* ethanol extract for industrial application as a natural material targeting itch-associated inflammation**

Kyeong Rok Kang[1], Seul Ah Lee[2], and Do Kyung Kim[1,*]

[1]*Department of Oral Physiology, College of Dentistry, Chosun University, Gwangju 61452, Republic of Korea*
[2]*Department of Biomaterials Development Research, L&K Co., Ltd, Gwangju, 61436, Republic of Korea*

P-04 **Isolation of Rare Actinomycetes and Discovery of Novel Natural Products Using an iChip-Inspired Cultivation Approach**

Hee-Tae Yeo[1,2], Na Yul Woo[1,3], Sooyeon Lee[1], Jin-Soo Park[1,2*]

[1]*Natural Product Systems Biology Center, Korea Institute of Science and Technology, Gangneung 25451, Republic of Korea*
[2]*Natural Product Applied Science, KIST School, University of Science and Technology, Gangneung 25451, Republic of Korea*
[3]*Department of Biology, Kangwon National University, Gangneung 25457, Republic of Korea*

P-05 **Serum Metabolite Profiling in a Clinical Trial of Black Ginseng for Liver Function Improvement: An NMR and GC-TOF/MS Based Study**

Woo-Cheol Shin[1], Jin-Kyu Jang[1], Hyeon Seon Na[1], Bo-Ram Choi[2], Young-Seob Lee[2], and Dae Young Lee[1*]

[1]*BK21 FOUR KNU Creative BioResearch Group, School of Life Sciences, Kyungpook National University, Daegu 41566, Republic of Korea*
[2]*Department of Herbal Crop Research, National Institute of Horticultural and Herbal Science, RDA, Eumseong 27709, Republic of Korea*

P-06 **Protective Effects of Black Ginseng Extract Against Ethanol-Induced Hepatic Injury and the Potential Role of Ginsenoside Rh1: *In Vivo, In Vitro*, and *In Silico* Studies**

Hyeon Seon Na[1], Jeon Hwang-Bo[2], Woo-Cheol Shin[1], Jin-Kyu Jang[1], Bo-Ram Choi[3] and Dae Young Lee[1*]

[1]*BK21 FOUR KNU Creative BioResearch Group, School of Life Sciences, Kyungpook National University, Daegu 41566, Republic of Korea*
[2]*Department of Genetic Engineering, Graduate School of Biotechnology, Kyung Hee University, Yongin 17104, Republic of Korea*
[3]*Department of Herbal Crop Research, National Institute of Horticultural and Herbal Science, Rural Development Administration, Eumseong 27709, Republic of Korea*

P-07 **Integrated Chemical Profiling, Metabolomics, and Network Pharmacology of *Platycodon grandiflorum* Extract against Ethanol-Induced Liver Injury**

Jin-Kyu Jang[1], Woo-Cheol Shin[1], Hyeon Seon Na[1], Dahye Yoon[2] and Dae Young Lee[1*]

[1]*BK21 FOUR KNU Creative BioResearch Group, School of Life Sciences, Kyungpook National University, Daegu 41566, Republic of Korea*

[2]*Department of Herbal Crop Research, National Institute of Horticultural and Herbal Science, RDA, Eumseong 27709, Republic of Korea*

P-08 **Network Pharmacology-Guided Evaluation of *Phedimus aizoon* Flavonoids in BV2 Microglial Cells**

Yong Jae Jeong[1], Jin-Kyu Jang[1], Woo-Cheol Shin[1], Hyeon Seon Na[1], Jeong Hyun Park[1], Dong-Sung Lee[2] and Dae Young Lee[1*]

[1]*BK21 FOUR KNU Creative BioResearch Group, School of Life Sciences, Kyungpook National University, Daegu 41566, Republic of Korea*

[2]*Research Institute of Pharmaceutical Sciences, College of Pharmacy, Chosun University, Gwangju 61452, Republic of Korea*

P-09 **Isolation and Identification of Compounds from *Schisandra chinensis* Pomace and Network Pharmacology-Based Evaluation of Their Potential Biological Activities**

Gun Woo Kim, Hyeon Seon Na, Seo Yeon Lee, Young geon Han and Dae Young Lee[*]

BK21 FOUR KNU Creative BioResearch Group, School of Life Sciences, Kyungpook National University, Daegu 41566, Republic of Korea

P-10 **An Integrated Approach to the Extraction, Fractionation and Structural Characterization of Acidic Polysaccharides from Processed Korean Ginseng**

Seo Yeon Lee, Jin-Kyu Jang, Young Geon Han, Gun Woo Kim, Won Jin Choi and Dae Young Lee[*]

BK21 FOUR KNU Creative BioResearch Group, School of Life Sciences, Kyungpook National University, Daegu 41566, Republic of Korea

P-11 **Structural Elucidation of Isolated Metabolites and Comprehensive profiling of Mulberry fruit Across Maturation Stages**

Young geon Han, Woo-Cheol Shin, Gun woo Kim, Seo Yeon Lee, Sin Hee Park and Dae Young Lee[*]

BK21 FOUR KNU Creative BioResearch Group, School of Life Sciences, Kyungpook National University, Daegu 41566, Republic of Korea

P-12 **Apoptosis and Autophagy Inducing Effects of a New Quinone Compound on HCT116 Colorectal Cancer Cells**

Ye Jin Kim[1], Hyun Bong Park[1*]

[1]*Department of Biology, Kangwon National University, Gangneung 25457, Republic of Korea*

P-13 **Structure elucidation and in-silico study of megastigmane glycosides from the extract of *Ardisia humilis* Vahl leaves to PPARs receptor for the treatment of metabolic syndrome**

Alfan Danny Arbianto[1,2,3], Seon Min Oh[1], Hyoung-Geun Kim[1], Jongmin Ahn[1], Kyujong Lee[4], Ngatinem Ngatinem[3], Agung Eru Wibowo[3], Jin-Hyub Paik[4], Hyung Won Ryu[1], Sofa Fajriah[3], Sangho Choi[4], and Sei-Ryang Oh[1,2*]

[1]*Natural Product Research Center, Korea Research Institute of Bioscience and Biotechnology*

[2]*KRIBB-SCHOOL, University of Science and Technology*

[3]*Research Center for Pharmaceutical Ingredients and Traditional Medicine, BRIN*

[4]*International Biological Material Research Center, KRIBB*

P-14 **Phytochemical Profiling and Analytical Method Validation for the Quality Assessment of *Ardisia humilis* Vahl**

Alfan Danny Arbianto[1,2,3], Seon Min Oh[1], Jongmin Ahn[1], Kyujong Lee[4], Ngatinem Ngatinem[3], Agung Eru Wibowo[3], Jin-Hyub Paik[4], Hyung Won Ryu[1], Sofa Fajriah[3], Sangho Choi[4], and Sei-Ryang Oh[1,2*]

[1]*Natural Product Research Center, Korea Research Institute of Bioscience and Biotechnology*
[2]*KRIBB-SCHOOL, University of Science and Technology*
[3]*Research Center for Pharmaceutical Ingredients and Traditional Medicine, BRIN*
[4]*International Biological Material Research Center, KRIBB*

P-15 **Metabolomic signatures of functional phytochemical diversity in spinach cultivars**

Su-Yeon Lee[1,2], Hyoung-Geun Kim[1], Doo-Young Kim[1], Su-Ah Lee[1], Seul Beom Yu[1], Jongmin Ahn[1], Sei-Ryang-Oh[1], Eun Kyoung Seo[2*], Hyung Won Ryu[1*]

[1]*Natural Product Research Center and Natural Product Central Bank, KRIBB, 30-Yeongudanji-ro, Ochang-eup, Cheongwon-gu, Cheongju-si, Chungcheongbuk-do, 28116, Korea*
[2]*College of Pharmacy, Graduate School of Pharmaceutical Sciences, Ewha Womans University, 52, Ewhyeodae-gil, Seodaemun-gu, Seoul, 03760, Korea*

P-16 **Seasonal metabolomic variation and anticancer potential of bioactive metabolites in *Paulownia tomentosa* fruits**

Su-Yeon Lee[1,2†], In-Seo Heo[1,2†], Hyoung-Geun Kim[1], Mun-Ock Kim[1], Inhyeok Song[3], Jaeha Kang[3], Jaewon Yang[3], Eun-Bin Kwon[1], Myung-Ji Kang[1], Doo-Young Kim[1], Jung-Hee Kim[1], Sei-Ryang-Oh[1], Daesik Jeong[4*], Eun Kyoung Seo[2*], Hyung Won Ryu[1*]

[1]*Natural Product Research Center and Natural Product Central Bank, KRIBB, 30-Yeongudanji-ro, Ochang-eup, Cheongwon-gu, Cheongju-si, Chungcheongbuk-do, 28116, Korea*
[2]*College of Pharmacy, Graduate School of Pharmaceutical Sciences, Ewha Womans University, 52, Ewhyeodae-gil, Seodaemun-gu, Seoul, 03760, Korea*
[3]*Bio Dx Group, InSiliCox, 202-ho, 42 Bongeunsa-ro 30-gil, Gangnam-gu, Seoul 06136, Korea*
[4]*Faculty of Software Convergence, Sangmyung University, 20 Hongjimun 2-gil, Jongno-gu, Seoul 03016, Korea*
[†]*These authors have equally contributed as the first author*

P-17 **Comparative metabolomic profiling of *Lysimachia mauritiana* Lam. across cultivation environments: Differentially abundant metabolites and environment-specific biomarkers**

Su-Ah Lee[1,2], Hyoung-Geun Kim[1], Hangyeol Lee[1], Doo-Young Kim[1,] Su-Yeon Lee[1,2], Seul Beom Yu[1], Soo Ho Yeon[3], Hyoung Tae Kim[4], Jin-Hyub Paik[5], Sei-Ryang Oh[1], Eun Kyoung Seo[2*], Hyung Won Ryu[1*]

[1]*Natural Product Research Center and Natural Product Central Bank, KRIBB, 30-Yeongudanji-ro, Ochang-eup, Cheongwon-gu, Cheongju-si, Chungbuk, 28116, Republic of Korea*
[2]*College of Pharmacy, Graduate School of Pharmaceutical Sciences, Ewha Womans University, 52, Ewhyeodae-gil, Seodaemun-gu, Seoul, 03760, Korea*
[3]*Kolmar BNH, 6-Hanbang expo-ro, Jecheon-si, Chungcheongbuk-do, 27116, Republic of Korea*
[4]*College of Ecology and Environmental Science, Kyungpook National University, 2559, Gyeongsang-daero, Sangju, Gyeongsangbuk-do, 37224, Korea*
[5]*International Biological Material Research Center, KRIBB, 125-Gwahak-ro, Yuseong-gu, Daejeon, 34141, Republic of Korea*

P-18 **Identification of tissue-specific chemical markers and antioxidant activities in different plant parts of *Vitex rotundifolia* L. f. using UPLC-QTOF/MS-based metabolomics**

Seul Beom Yu[1,2], Hyoung-Geun Kim[1],Hangyeol Lee[1], Doo-Young Kim[1], In Soo Suh[3], Sung Chun Kim[3],Sei-Ryang Oh[1], Dongho Lee[2*],Hyung Won Ryu[1*]

[1]*Natural Product Research Center, Korea Research Institute of Bioscience and Biotechnology KRIBB, 30-Yeongudanji-ro, Ochang-eup, Cheongwon-gu, Cheongju-si, Chungbuk, 28116, Republic of Korea*
[2]*Department of Plant Biotechnology, College of Life Sciences and Biotechnology, Korea University, 145-Anam-ro, Seongbuk-gu, Seoul, 02841, Republic of Korea*
[3]*Biodiversity Research Institute, Clean Bio Business Division, JejuTechnopark, 338, Sinryedong-ro, Namwon-eup, Seogwipo-si, Jeju-do, 63608, Republic of Korea*

P-19 **Relationship between Chemical Composition and Biological Activity of Supercritical Fluid Extracted Arthropod Biomass**

Won Jun Lee*

Ecological Technology Research Team, National Institute of Ecology, Seocheon-gun 33657, Republic of Korea

P-20 **Contribution of Minor Coumarins to the Anti-inflammatory Activity of *Cnidium monnieri* Assessed by Knock-out Fractionation**

Sangmin Lee[1,2], Sun Yong Park[2], Yerim Joo[1,2], Eunbeen Shin[1,2] and Seon Beom Kim[1,2,3]*

[1]*Department of Food Science and Technology, College of Natural Resources and Life Science, Pusan National University, Miryang 50463, Republic of Korea*

[2]*Institute for Future Earth, Pusan National University, Busan 46241, Republic of Korea*

[3]*Food Tech Innovation Center, Life and Industry Convergence Research Institue, Pusan national University, Miryang, Republic of Korea*

P-21 **Chemical Characterization of Bioactive Sulfated Polysaccharides Isolated from *Stichopus japonicus***

Eunbeen Shin[1,2], Ibukunoluwa Fola Olawuyi[2], Sangmin Lee[1,2], Minhyuk Jeong[1,2], and Seon Beom Kim[1,2,3,*]

[1]*Department of Food Science and Technology, College of Natural Resources and Life Science, Pusan National University, Miryang 50463, South Korea*

[2]*Institute for Future Earth, Pusan National University, Busan 46241, South Korea*

[3]*Food Tech Innovation Center, Life and Industry Convergence Research Institute, Pusan National University, Miryang, South Korea*

P-22 **Rapid Screening of Binding Constituents in Blueberry Extract Using Tyrosinase**

Dayeon Kang[1], **Hyoung-Geun Kim**[2], and Hyung Won Ryu[2*]

[1]*Osong High School, Cheongju, Chungbuk, 28165, Republic of Korea*

[2]*Natural Product Research Center, Korea Research Institute of Bioscience and Biotechnology KRIBB, 30-Yeongudanji-ro, Ochang-eup, Cheongwon-gu, Cheongju-si, Chungbuk, 28116, Republic of Korea*

P-23 **Identification of Metabolites and Gene Expression Changes in *Vigna angularis* L. Sprouts under Indoor and Outdoor Cultivation Conditions Using HPLC-QTOF-MS and RT-PCR**

Bo Gun Jeong[1], **Han gyeol Lee**[2], and Hyung Won Ryu[2*]

[1]*Cheongwon High School, 9 Guryong 5-gil, Ochang-eup, Cheongwon-gu, Cheongju 28121, Republic of Korea*

[2]*Natural Product Research Center and Natural Product Central Bank, KRIBB, 30-Yeongudanji-ro, Ochang-eup, Cheongwon-gu, Cheongju-si, Chungcheongbuk-do 28116, Republic of Korea*

P-24 **A Study on Functional Materialization Using Persimmon-Peel Biomass and the Development of Cosmetics: Optimization of Enzymatic Fermentation and Ultra-Fine Milling Extraction Processes and Efficacy Evaluation**

Yu-Rim Lee, Mi-Ae Do, Hwa-Jin Suh*

Convergence R&D Center, Korea Textile Smart Process Institute, Gongju 31156, Republic of Korea

P-25 **Anti-inflammatory Activity of Cheongyeolhaedok-tang via the Regulation of Inflammatory Chemokines and Antimicrobial Peptides in a Psoriasis-Like Human Keratinocyte Model**

[1]**Tae-hee Go**, [1]Na-Yeon Kang, [2]Yeop Jung, [2]Sun-Young Park, [1]Hyung-Seo Hwang*

[1]*Department of Bio-Cosmetic Science, Semyung University, Jecheon, Chungbuk 27136, South Korea*

[2]*Department of Physiology, College of Korean Medicine, Semyung University, Jecheon, Chungbuk 27136, South Korea*

P-26 **Protective Effects of *Geranium thunbergii* Extract Against UV-Induced Skin Aging in Human Dermal Fibroblasts and Zebrafish**

Ye-Jin Lee[1,3], Bongkyun Park[1,3], Heung Joo Yuk[2,3] and Young Sook Kim[1,3,*]

[1]Korean Medicine Convergence Research Division, Korea Institute of Oriental Medicine, Daejeon 34054, Republic of Korea
[2]NeuroGlymph Imaging and Modulation Center, Korea Institute of Oriental Medicine, Daejeon 34054, Republic of Korea
[3]School of Korean Convergence Medical Science, University of Science and Technology (UST), Daejeon, 34113, Republic of Korea

P-27 **Screening of Psoriasis-Related Candidate Materials from Vietnamese Medicinal Plants Based on Antioxidant Activity and Total Phenolic Content**

Jae Jun Kim, Yeon Woong Jung, Hyung Seo Hwang*

Department of Bio-Cosmetic Science, Semyung University, Jecheon, Chungbuk 27136, South Korea

P-28 **Isolation and Structural Characterization of Khellactone-Type Coumarins from *Peucedanum japonicum* with Potential Anti-Neuroinflammatory Activity**

Jeong Hyun Park[1], Woo-Cheol Shin[1], Jin-Kyu Jang[1], Hyeon Seon Na[1], Yong Jae Jeong[1], Dong-Sung Lee[2] and Dae Young Lee[1*]

[1]BK21 FOUR KNU Creative BioResearch Group, School of Life Sciences, Kyungpook National University, Daegu 41566, Republic of Korea
[2]Research Institute of Pharmaceutical Sciences, College of Pharmacy, Chosun University, Gwangju 61452, Republic of Korea

P-29 **Regional Metabolomic Differentiation of *Perilla frutescens* Leaves Based on LC-Q-TOF/MS and Multivariate Analysis**

Na Rae Kang[a,b], Yun Gon Son[a,b], Seungjae Jang[a,b], Seungyu Lee[a,b] and Jeong Yoon Kim[a,b,*]

[a]Department of Pharmaceutical Engineering, ABC-RLRC, IALS, Gyeongsang National University, Jinju, 52725, Republic of Korea
[b]Gyeongnam Aerospace & Defense Institute of Science and Technology, Gyeongsang National University, Jinju 52828, Republic of Korea

P-30 **Tyrosinase Inhibitory Activity of Compounds Isolated from *Agastache rugosa* Seeds as Potential Natural Skin-Whitening Agents**

Seungjae Jang[a], Seungyu Lee[a], Hoil Kim[a], Jeong Yoon Kim[a,*]

[a]Department of Pharmaceutical Engineering, ABC-RLRC, IALS, Gyeongsang National University, Jinju, 52725, Republic of Korea

P-31 **Anti-biofilm Potential of Furanocoumarins Isolated from *Peucedanum japonicum* Seeds against *Pseudomonas aeruginosa***

Jihyeon Choi[a], Hoil Kim[a], Jeong Yoon Kim[a,*]

[a]Department of Pharmaceutical Engineering, ABC-RLRC, IALS, Gyeongsang National University, Jinju, 52725, Republic of Korea

P-32 **Tyrosinase Inhibition and Kinetic Characterization of Tracheloside and Arctiin from *Synurus deltoides* Seeds**

Bashir Shanza[a], Seungmin Oh[a], Juhyeon Kim[a], Jeong Yoon Kim[a,*]

[a]Department of Pharmaceutical Engineering, ABC-RLRC, IALS, Gyeongsang National University, Jinju, 52725, Republic of Korea

P-01

Lancemaside A isolated from *Codonopsis lanceolata* ameliorates inflammation-induced intestinal barrier dysfunction

Hyeon-Jeong Jeong[1], So-Ri Son[2], Dae Sik Jang[2] and Hyun Bong Park[1,*]

[1]*Department of Biological Sciences, Kangwon National University, Gangneung 25457, Republic of Korea*
[2]*Department of Biomedical and Pharmaceutical Sciences, Graduate School, Kyung Hee University, Seoul 02447, Republic of Korea*

Codonopsis lanceolata is an edible plant traditionally consumed as both a food and herbal medicine, yet the biological activities of its aerial parts remain largely unexplored. Compounds isolated from the aerial parts were screened for anti-inflammatory activity, with Lancemaside A showing the strongest activity among the tested compounds. Lancemaside A reduced inflammatory responses in intestinal epithelial cells and improved barrier integrity, as shown by increased transepithelial electrical resistance (TEER), decreased paracellular permeability, and recovery of tight junction protein expression. Transcriptomic analysis showed changes in pathways related to epithelial barrier function and antioxidant responses, along with increased expression of tight junction- and Nrf2-associated genes. ARE-luciferase and protein expression analyses further showed that Lancemaside A activates Nrf2-associated signaling. These results suggest that Lancemaside A is an active constituent of *C. lanceolata* aerial parts with intestinal barrier-protective properties and potential as a food-derived functional ingredient for intestinal health.

Keywords *Codonopsis lanceolata*, lancemaside A, inflammation, intestinal barrier, tight junction, Nrf2

P-02

Standardization and Biological Activity Screening of *Cakile edentula* (Bigelow) Hook. Leaf Extract

Jun Seok Wang[1], Chul Woong Ho[1], Hyeon-Jeong Jeong[1], Ye Jin Kim[1], Aram Kim[1], Kyung-Won Min[1], and Hyun Bong Park[1*]

[1]*Department of Biological Sciences, College of Natural Sciences, Kangwon National University, Gangneung, 25457, Republic of Korea*

Natural products have gained increasing attention as important sources of bioactive compounds for disease prevention and therapeutic applications. However, the biological activity and phytochemical composition of *Cakile edentula*, a halophytic plant inhabiting coastal dunes, remain largely unexplored. In this study, we evaluated the antioxidant and anti-inflammatory properties of *C. edentula* leaf extract and investigated its potential to modulate therapy-induced senescence (TIS) in cancer cells. The extract exhibited free radical scavenging activity and suppressed inflammatory responses by reducing nitric oxide production and the expression of pro-inflammatory cytokines. Given that oxidative stress and inflammatory signaling are key drivers of TIS, we hypothesized that the extract may influence this process. To test this, a doxorubicin-induced senescence model was established. *C. edentula* extract significantly reduced senescence-associated β-galactosidase (SA-β-gal) activity and decreased *p21* expression, indicating attenuation of senescence-associated phenotypes. Importantly, the extract markedly suppressed the expression of senescence-associated secretory phenotype (SASP) factors, including *IL-6*, *IL-8*, and *TGF-β*, suggesting that it modulates the inflammatory and pathological features of TIS rather than simply inhibiting senescence itself. Furthermore, phytochemical analysis identified various bioactive compounds, including flavonoids and fatty acid derivatives, and characterized the structures of nine major constituents. A marker compound was selected and quantitatively analyzed, supporting the feasibility of extract standardization. Collectively, these findings demonstrate that *C. edentula* leaf extract attenuates the pathological features of therapy-induced senescence through its antioxidant and anti-inflammatory activities. These results highlight its potential as a functional natural resource for modulating cancer-associated senescence and support its further development for therapeutic applications.

Keywords Natural Products, Phytochemicals, *Cakile edentula* (Bigelow) Hook., Brassicaceae, Standardization, Flavonoids, Antioxidant, Anti-inflammation

P-03

Bioactive potential of *Codium fragile* ethanol extract for industrial application as a natural material targeting itch-associated inflammation

Kyeong Rok Kang[1], Seul Ah Lee[2], and Do Kyung Kim[1,*]

[1]*Department of Oral Physiology, College of Dentistry, Chosun University, Gwangju 61452, Republic of Korea*
[2]*Department of Biomaterials Development Research, L&K Co., Ltd, Gwangju, 61436, Republic of Korea*

Marine-derived natural products have attracted increasing attention as sustainable sources of bioactive compounds with potential applications in functional foods, cosmetics, and related industries. *Codium fragile* (*C. fragile*), an edible green macroalga widely distributed in coastal regions of East Asia, has been reported to exhibit various biological activities, including antioxidant, anti-inflammatory, immunomodulatory, and skin-protective effects. However, its potential activity against itch-associated mast cell responses and the integrated regulation of inflammation and epidermal responses has not been fully characterized.

In this study, we investigated the potential itch-relieving and anti-inflammatory effects of a 70% ethanol extract of *C. fragile* (CF70E) using RBL-2H3 mast cells, RAW264.7 macrophages, and HaCaT keratinocytes. CF70E showed no significant cytotoxicity at the tested concentrations. In stimulated RBL-2H3 cells, CF70E significantly inhibited β-hexosaminidase release and reduced TNF-α and IL-6 production. These effects were accompanied by suppression of Syk and PLCγ1 phosphorylation, indicating attenuation of mast cell activation and degranulation-related signaling. In LPS-stimulated RAW264.7 macrophages, CF70E reduced nitric oxide, prostaglandin E2, TNF-α, and IL-6 production, together with decreased expression of iNOS and COX-2. In stimulated HaCaT keratinocytes, CF70E decreased the expression of inflammatory mediators, including MDC, TARC, and IL-1β, while modulating the expression of epidermal barrier-associated proteins such as filaggrin and involucrin. In addition, CF70E attenuated MAPK activation in RBL-2H3 and RAW264.7 cells.

Taken together, these findings demonstrate that CF70E exerts multiple biological activities related to mast cell degranulation, inflammatory regulation, and epidermal homeostasis. The present results support the potential utilization of C. fragile as a marine-derived natural functional material for the development of products targeting itch-associated inflammatory skin conditions and provide a biological basis for the industrial application of this underutilized marine resource

Keywords Codium fragile, Itch-relief, Anti-inflammatoty, β-hexosaminidase

P-04

Isolation of Rare Actinomycetes and Discovery of Novel Natural Products Using an iChip-Inspired Cultivation Approach

Hee-Tae Yeo[1,2], Na Yul Woo[1,3], Sooyeon Lee[1], Jin-Soo Park[1,2*]

[1]*Natural Product Systems Biology Center, Korea Institute of Science and Technology, Gangneung 25451, Republic of Korea*
[2]*Natural Product Applied Science, KIST School, University of Science and Technology, Gangneung 25451, Republic of Korea*
[3]*Department of Biology, Kangwon National University, Gangneung 25457, Republic of Korea*

Microbial secondary metabolites remain an invaluable source of therapeutic lead compounds, yet conventional laboratory cultivation limits access to the vast chemical diversity of uncultured microbes. To overcome rediscovery challenges and access specialized metabolites from fastidious microbes, we implemented a modified cultivation approach based on the *in situ* cultivation (iChip). Cell suspensions prepared from the aerial parts, belowground parts, and rhizosphere soil of *Calanthe coreana* Nakai were mixed with 1/10-strength YME agar, dispensed into 96-well filter plates, and incubated for 8 weeks either standardly or on top of soil collected from the *C. coreana* cultivation site. In the absence of soil exposure, 14 bacterial strains, including 6 actinomycetes, were isolated, whereas incubation on cultivation-site soil yielded 18 bacterial strains, including 14 actinomycetes, and 10 fungal strains. Notably, the number of actinomycete isolates increased from 6 to 14 following incubation on cultivation-site soil, and diverse actinomycete genera, including *Streptomyces*, *Actinacidiphila*, *Embleya*, and *Geodermatophilus*, were recovered. Among these isolates, an *Embleya* sp. strain was selected for metabolite profiling. LC-MS and HR-MS analyses revealed 8 major metabolite peaks, 6 of which were identified as known compounds, whereas 2 were distinguished from previously reported compounds. The two metabolites exhibited UV absorption characteristics consistent with a salicylic acid-derived chromophore and MS/MS fragmentation patterns similar to those of catenulobactins A and B, suggesting that they represent novel derivatives of the catenulobactin family. The compounds were subsequently isolated and purified by preparative HPLC, and structural elucidation using 1D and 2D NMR spectroscopy is ongoing. These finding demonstrate that an iChip-inspired *in situ* simulation approach effectively facilitates the isolation of previously inaccessible rare actinomycetes, serving as powerful platform for discovery novel natural products.

Keywords modified *in situ* cultivation, *Calanthe coreana*, rare actinomycetes, *Embleya*, microbial secondary metabolites, catenulobactin, metabolite profiling

P-05

Serum Metabolite Profiling in a Clinical Trial of Black Ginseng for Liver Function Improvement: An NMR and GC-TOF/MS Based Study

Woo-Cheol Shin[1], Jin-Kyu Jang[1], Hyeon Seon Na[1], Bo-Ram Choi[2], Young-Seob Lee[2], and Dae Young Lee[1*]

[1]BK21 FOUR KNU Creative BioResearch Group, School of Life Sciences, Kyungpook National University, Daegu 41566, Republic of Korea

[2]Department of Herbal Crop Research, National Institute of Horticultural and Herbal Science, RDA, Eumseong 27709, Republic of Korea

This study aims to characterize changes in the serum metabolome following black ginseng supplementation and explore their associations with clinical indicators of liver function. Serum samples were obtained from a 12-week, multicenter, randomized, double-blind, placebo-controlled clinical trial evaluating the efficacy and safety of black ginseng for improving liver function. Samples collected from the placebo and black ginseng groups at V2 (week 2) and V4 (week 4) are being analyzed by nuclear magnetic resonance (NMR) spectroscopy and gas chromatography–time-of-flight mass spectrometry (GC-TOF/MS). For NMR analysis, serum samples undergo protein removal and buffer adjustment. Separate aliquots undergo organic solvent extraction and derivatization for GC-TOF/MS analysis. Endogenous metabolites, including amino acids, organic acids, and sugars, will be identified or putatively annotated using authentic standards and spectral libraries. After quality control and normalization, metabolite levels will be compared between groups and across visits using multivariate and univariate analyses that account for repeated measurements within participants. Associations between changes in candidate metabolites and changes in clinical indicators of liver function and antioxidant markers will also be explored.

This study may provide insights into changes in serum metabolites following black ginseng supplementation and their potential relationship with liver function.

Keywords Black ginseng, clinical trial, human serum, NMR, GC-TOF/MS

P-06

Protective Effects of Black Ginseng Extract Against Ethanol-Induced Hepatic Injury and the Potential Role of Ginsenoside Rh1: *In Vivo, In Vitro*, and *In Silico* Studies

Hyeon Seon Na[1], Jeon Hwang-Bo[2], Woo-Cheol Shin[1], Jin-Kyu Jang[1], Bo-Ram Choi[3] and Dae Young Lee[1*]

[1]*BK21 FOUR KNU Creative BioResearch Group, School of Life Sciences, Kyungpook National University, Daegu 41566, Republic of Korea*

[2]*Department of Genetic Engineering, Graduate School of Biotechnology, Kyung Hee University, Yongin 17104, Republic of Korea*

[3]*Department of Herbal Crop Research, National Institute of Horticultural and Herbal Science, Rural Development Administration, Eumseong 27709, Republic of Korea*

Alcohol-induced liver damage (AILD), associated with chronic alcohol consumption, is characterized by oxidative stress and inflammation and remains a major health concern. Black ginseng, produced through repeated thermal processing of ginseng, has been reported to exhibit various pharmacological activities. However, its protective effects against AILD and the underlying mechanisms remain unclear. This study investigated the protective effects of black ginseng extract (BGE) against AILD and the potential molecular role of ginsenoside Rh1 using *in vivo*, *in vitro*, and *in silico* approaches. In ethanol-treated mice, BGE significantly reduced serum AST and ALT levels and increased hepatic catalase (CAT) and superoxide dismutase (SOD) activities. In HepG2 cells, BGE inhibited nitric oxide production, suppressed TNF-α, iNOS, COX-2, and IL-6 expression, and increased HO-1 expression. Ginsenoside Rh1 was quantified in BGE using UPLC-Q-TRAP-MS/MS. Network pharmacology and molecular docking analyses identified potential targets, signaling pathways, and multi-target interactions associated with Rh1.

Collectively, these findings suggest that BGE and ginsenoside Rh1 exert hepatoprotective effects by modulating oxidative stress and inflammatory signaling pathways. Rh1 may also serve as a quality control marker of BGE, supporting the potential application of BGE for the prevention and treatment of AILD.

Keywords Black ginseng, Ginsenoside Rh1, Alcohol-induced liver damage, Oxidative stress, Inflammation, Network pharmacology

P-07

Integrated Chemical Profiling, Metabolomics, and Network Pharmacology of *Platycodon grandiflorum* Extract against Ethanol-Induced Liver Injury

Jin-Kyu Jang[1], Woo-Cheol Shin[1], Hyeon Seon Na[1], Dahye Yoon[2] and Dae Young Lee[1*]

[1]BK21 FOUR KNU Creative BioResearch Group, School of Life Sciences, Kyungpook National University, Daegu 41566, Republic of Korea

[2]Department of Herbal Crop Research, National Institute of Horticultural and Herbal Science, RDA, Eumseong 27709, Republic of Korea

This study investigated the effects of *Platycodon grandiflorum* extract (PGE) on ethanol-induced liver injury using UPLC-Q-TOF/MS chemical profiling, NMR-based metabolite analysis, and network pharmacology

Deapi-platycodin D3, a triterpenoid saponin, was identified as a characteristic constituent of PGE. In mice with ethanol-induced liver injury, PGE (50–200 mg/kg) reduced serum AST and ALT levels, improved hepatic antioxidant measures, and alleviated lipid accumulation and histopathological damage. NMR-based metabolomic analysis showed partial attenuation of ethanol-associated metabolic changes.

In ethanol-treated HepG2 cells, PGE improved cell viability, decreased NO production, suppressed the expression of iNOS, COX-2, IL-6, and TNF-α, and increased HO-1 expression. Deapi-platycodin D3 also attenuated ethanol-induced cytotoxicity. Network pharmacology analysis indicated that the predicted targets associated with deapi-platycodin D3 were involved in inflammation- and oxidative stress-related pathways. These predictions were broadly consistent with the biological responses observed in the experimental models.

These findings suggest that PGE may alleviate ethanol-induced liver injury, with deapi-platycodin D3 as a potential contributing constituent.

Keywords *Platycodon grandiflorum,* Deapi-platycodin D3, NMR-based metabolomics, Network pharmacology, Alcoholic liver injury

P-08

Network Pharmacology-Guided Evaluation of *Phedimus aizoon* Flavonoids in BV2 Microglial Cells

Yong Jae Jeong[1], Jin-Kyu Jang[1], Woo-Cheol Shin[1], Hyeon Seon Na[1], Jeong Hyun Park[1], Dong-Sung Lee[2] and Dae Young Lee[1*]

[1]*BK21 FOUR KNU Creative BioResearch Group, School of Life Sciences, Kyungpook National University, Daegu 41566, Republic of Korea*
[2]*Research Institute of Pharmaceutical Sciences, College of Pharmacy, Chosun University, Gwangju 61452, Republic of Korea*

Phedimus aizoon is a medicinal plant containing phenolic compounds and flavonoid glycosides. This study isolated flavonoids from *P. aizoon* and investigated their potential relevance to neuroinflammation using network pharmacology, molecular docking, and a cellular assay.

Compounds were isolated by open-column chromatography and medium-pressure liquid chromatography. Their molecular masses were examined by UPLC-Q-TOF/MS, and their structures were elucidated by NMR spectroscopy. Six flavonoids were identified as myricitrin, quercitrin, 2″-O-galloylmyricitrin, 3″-O-galloylmyricitrin, 2″-O-galloylquercitrin, and 3″-O-galloylquercitrin.

Network pharmacology analysis was conducted to identify potential targets and biological pathways associated with neuroinflammation. Protein–protein interaction analyses identified eight predicted hub genes associated with neuroinflammation-related pathways. Molecular docking was used to predict interactions between the isolated compounds and selected target proteins. The effects of the compounds on cell viability and nitric oxide production were then evaluated in lipopolysaccharide-stimulated BV2 microglial cell.

Collectively, These findings provide a basis for further investigation of *P. aizoon* flavonoids as potential modulators of neuroinflammatory responses.

Keywords *Phedimus aizoon*, Neuroinflammation, Flavonoids, Network pharmacology

P-09

Isolation and Identification of Compounds from *Schisandra chinensis* Pomace and Network Pharmacology-Based Evaluation of Their Potential Biological Activities

Gun Woo Kim, Hyeon Seon Na, Seo Yeon Lee, Young geon Han and Dae Young Lee*

BK21 FOUR KNU Creative BioResearch Group, School of Life Sciences, Kyungpook National University, Daegu 41566, Republic of Korea

Schisandra chinensis fruit is widely used in food ingredient and herbal medicine, but its processing generates pomace that may retain bioactive constituents. This study aimed to isolate and identify compounds from *S. chinensis* pomace (SCP) and explore their potential biological relevance using network pharmacology.

SCP was extracted with methanol and partitioned into EtOAc, *n*-BuOH, and aqueous fractions. Compounds were isolated from the EtOAc fraction by repeated open-column chromatography. Their structures were determined using 1D and 2D NMR spectroscopy, supported by accurate-mass data from UPLC-Q-TOF/MS. Network pharmacology was used to predict potential protein targets and associated pathways, and selected compound–target interactions were assessed by molecular docking.

In the network pharmacology analysis, GO and KEGG enrichment analyses revealed functions and pathways associated with the predicted targets, while protein-proein interaction (PPI) network and topology analyses identified key target proteins. Neuroprotection-related pathways and proteins were identified, suggesting that compounds isolated from SCP may have neuroprotective potential.

Overall, this study is expected to contribute to the reevaluation of SCP and provide fundamental information for its potential use as a functional food ingredient.

Keywords *Schisandra chinensis* pomace, Column chromatography, NMR, LC/MS, Network Pharmacology

P-10

An Integrated Approach to the Extraction, Fractionation and Structural Characterization of Acidic Polysaccharides from Processed Korean Ginseng

Seo Yeon Lee, Jin-Kyu Jang, Young Geon Han, Gun Woo Kim, Won Jin Choi and Dae Young Lee*

BK21 FOUR KNU Creative BioResearch Group, School of Life Sciences, Kyungpook National University, Daegu 41566, Republic of Korea

Acidic polysaccharides from Korean ginseng have attracted interest for their reported biological activities. However, their composition and structural properties vary according to processing and extraction conditions. This study aimed to compare polysaccharide fractions from white, red, and black ginseng and characterize fractions enriched in acidic polysaccharides.

Ginseng samples were pretreated with aqueous ethanol at different concentrations and subsequently extracted with hot water under reflux. The resulting crude polysaccharides were fractionated according to molecular weight using an Amicon ultrafiltration system, and the acidic polysaccharide content of each fraction was quantified using the carbazole–sulfuric acid assay. Neutral and acidic fractions were separated by DEAE-cellulose chromatography. The fraction most enriched in acidic polysaccharides was purified and desalted, followed by GC-TOF/MS monosaccharide profiling and NMR structural analysis.

This approach enables comparison of polysaccharide fractions across differently processed ginseng samples. Further compositional and structural analyses will help identify fractions for subsequent structure–activity studies.

Keywords Acidic polysaccharides, Processed ginseng, ion-exchange chromatography, GC-TOF/MS, Structural characterization, NMR spectroscopy

P-11

Structural Elucidation of Isolated Metabolites and Comprehensive profiling of Mulberry fruit Across Maturation Stages

Young geon Han, Woo-Cheol Shin, Gun woo Kim, Seo Yeon Lee, Sin Hee Park and Dae Young Lee*

BK21 FOUR KNU Creative BioResearch Group, School of Life Sciences, Kyungpook National University, Daegu 41566, Republic of Korea

Mulberry fruit (*Morus alba* L.) contains diverse bioactive constituents whose composition changes during maturation. This study investigated metabolite variation across fruit maturation stages and explored candidate chemical markers for quality evaluation.

In this study, immature and mature fruits were extracted and fractionated separately, and the resulting extracts were combined for chromatographic isolation of individual metabolites. The structures of the isolated compounds were elucidated using NMR spectroscopy, and an in-house metabolite library was established for subsequent metabolite profiling. For profiling, fruits were harvested at five maturation stages, and metabolite variations were analyzed using UPLC-Q-TOF/MS-based profiling combined with multivariate statistical analyses, including PCA and PLS-DA. The metabolomic analysis revealed distinct differences in chemical composition among maturation stages, indicating that specific metabolites are closely associated with fruit developmental status.

These results provide insight into maturation dependent metabolic changes in mulberry fruit and may contribute to the identification of chemical markers for quality control, standardization, and functional material development.

Keywords Mulberry, NMR, metabolite profiling, UPLC-Q-TOF/MS, multivariate statistical analysis

P-12

Apoptosis and Autophagy Inducing Effects of a New Quinone Compound on HCT116 Colorectal Cancer Cells

Ye Jin Kim[1], Hyun Bong Park[1*]

[1]*Department of Biology, Kangwon National University, Gangneung 25457, Republic of Korea*

Natural products are important resources for drug development because of their diverse biological activities and structural diversity, and they have contributed to anticancer agent development. Colorectal cancer continues to have high incidence and mortality worldwide, underscoring the ongoing need for new anticancer agents. In a previous study, a 1,4-naphthoquinone–indole conjugate metabolite derived from gut microbiota exhibited cytotoxic activity. Based on this compound, we synthesized amiquinone, a new quinone compound with stronger cytotoxic activity. In this study, we found that amiquinone induced apoptosis in human colorectal cancer HCT116 cells. It increased PARP and caspase-3 cleavage, while flow cytometry showed a larger population of apoptotic cells. RNA sequencing revealed higher *TNFRSF10B* expression, suggesting activation of the extrinsic apoptotic pathway. Transcriptomic profiling also identified changes in genes related to ER stress, MAPK signaling, oxidative stress, and autophagy. Autophagy involvement was supported by increased *MAP1LC3B* transcript levels, increased LC3B-I to LC3B-II conversion, and reduced p62 protein levels. Overall, this study is the first to show the cytotoxic activity of amiquinone in HCT116 cells and suggests that amiquinone induces apoptosis and autophagy. These findings provide insight into the cellular mechanisms underlying the cytotoxic activity of amiquinone, a new quinone compound derived from a gut microbiota metabolite.

Keywords 1,4-naphthoquinone, Microbial metabolites, Colorectal Cancer, apoptosis, autophagy

P-13

Structure elucidation and in-silico study of megastigmane glycosides from the extract of *Ardisia humilis* Vahl leaves to PPARs receptor for the treatment of metabolic syndrome

Alfan Danny Arbianto[1,2,3], Seon Min Oh[1], Hyoung-Geun Kim[1], Jongmin Ahn[1], Kyujong Lee[4], Ngatinem Ngatinem[3], Agung Eru Wibowo[3], Jin-Hyub Paik[4], Hyung Won Ryu[1], Sofa Fajriah[3], Sangho Choi[4], and Sei-Ryang Oh[1,2*]

[1]*Natural Product Research Center, Korea Research Institute of Bioscience and Biotechnology*
[2]*KRIBB-SCHOOL, University of Science and Technology*
[3]*Research Center for Pharmaceutical Ingredients and Traditional Medicine, BRIN*
[4]*International Biological Material Research Center, KRIBB*

Ardisia humilis Vahl. (AHV), known as Lampeni, which distributes widely across the tropical rainforests of Southeast Asia. This plant has traditionally been used in various forms of medicinal herbs on the Sumatra island in Indonesia. Previous phytochemical studies have revealed that AHV contains of secondary metabolites, including carotenoids, flavonoids, triterpenoids, and megastigmane derivatives. Megastigmanes are C13 compounds with a structure basically consisting of a backbone of six carbon atoms, with two methyl groups substituted at the C-1 position, a methyl group at the C-5 position, and four carbon atoms connected at the C-6 position. The glycosides attached to megastigmanes increase their water solubility, making them emerging bioactive plant compounds that help mitigate metabolic syndrome by suppressing chronic inflammation, reducing oxidative stress, and regulating lipid and glucose metabolism from natural products. Meanwhile, there are few specific studies focusing on the metabolic processes of these megastigman glycosides with PPARs receptors. In this study, megastigman glycosides were isolated from AHV followed by computer docking to evaluate their potential as PPARs receptor. The molecular docking results revealed that the lowest binding energies fell within the range of -6.1 to -12.3 kcal/mol with RMSD value of < 2.0. Therefore, this in-silico simulation study is crucial for serving as a guideline regarding the biological activity of the substances, and these findings can support further biological studies.

Keywords *Ardisia humilis*, in-silico, megastigmane glycosides

P-14

Phytochemical Profiling and Analytical Method Validation for the Quality Assessment of *Ardisia humilis* Vahl

Alfan Danny Arbianto[1,2,3], Seon Min Oh[1], Jongmin Ahn[1], Kyujong Lee[4], Ngatinem Ngatinem[3], Agung Eru Wibowo[3], Jin-Hyub Paik[4], Hyung Won Ryu[1], Sofa Fajriah[3], Sangho Choi[4], and Sei-Ryang Oh[1,2*]

[1]*Natural Product Research Center, Korea Research Institute of Bioscience and Biotechnology*
[2]*KRIBB-SCHOOL, University of Science and Technology*
[3]*Research Center for Pharmaceutical Ingredients and Traditional Medicine, BRIN*
[4]*International Biological Material Research Center, KRIBB*

Ardisia humilis Vahl (AHV) is one medicinal plant used in Indonesia jamu, popularly known as Lampeni which is widely distributed in two big islands in Sumatra and Java. It has been commonly used as traditional medicine to treat stomach and skin problems. AHV has been reported to be rich in phenolic and flavonoid content which has exhibited various biological activities including antioxidant. Jamu is often produced without any standard that is used to maintain the quality of the extract product, therefore jamu has inconsistency in efficacy and in the process of extract production with high flavonoid content. In this study, metabolomic-based screening was conducted to screen phytochemical composition of AHV to get flavonoid-rich fraction for jamu production followed by TLC and HPLC validation of the kaempferol as reference standard to control quality of AHV. The developed TLC effectively identified the kaempferol with a visible detection limit (LOD) of 25 ppm, making it suitable for rapid raw material verification. The linear regression analysis data for the calibration plots showed a good linear relationship over the concentration range of 1.56 – 200 µg/mL for kaempferol. The method was validated for linearity, precision, robustness and recovery according to ICH guidelines. Statistical analysis showed that the method was repeatable and reproducibility in various tested column types that can be applied for routine quality control of raw materials in industry.

Keywords *Ardisia humilis*, phytochemical, validation

P-15

Metabolomic signatures of functional phytochemical diversity in spinach cultivars

Su-Yeon Lee[1,2], Hyoung-Geun Kim[1], Doo-Young Kim[1], Su-Ah Lee[1], Seul Beom Yu[1], Jongmin Ahn[1], Sei-Ryang-Oh[1], Eun Kyoung Seo[2*], Hyung Won Ryu[1*]

[1]Natural Product Research Center and Natural Product Central Bank, KRIBB, 30-Yeongudanji-ro, Ochang-eup, Cheongwon-gu, Cheongju-si, Chungcheongbuk-do, 28116, Korea
[2]College of Pharmacy, Graduate School of Pharmaceutical Sciences, Ewha Womans University, 52, Ewhyeodae-gil, Seodaemun-gu, Seoul, 03760, Korea

Spinach (*Spinacia oleracea* L.) is a widely consumed leafy vegetable recognized for its nutritional value and diverse phytochemical composition. However, comprehensive information on cultivar-dependent variation in spinach metabolites remains limited. In this study, LC-MS-based untargeted metabolomics was employed to characterize phytochemical diversity among 15 spinach cultivars cultivated under the same conditions. Aerial tissues were extracted with 80% ethanol and analyzed using ultra-high-performance liquid chromatography coupled with quadrupole time-of-flight mass spectrometry (UPLC-QTOF/MS). Metabolites were tentatively identified based on accurate mass, molecular formula, and MS/MS fragmentation patterns, and multivariate statistical analyses were performed to investigate cultivar-associated metabolic variation.

A total of 37 metabolites were profiled, including phenolic acids, flavonoids and flavonoid glycosides, coumarins, and triterpenoid saponins. In particular, diverse spinacetin- and patuletin-derived flavonoid glycosides and a series of spinach saponins, including spinaciasaponins A–E, celosin I, and medicagenic acid- and yunganogenin K-type saponins, were detected. Principal component analysis revealed distinct metabolic patterns among the 15 cultivars, and loading plots and comparative analyses identified metabolites that contributed to cultivar discrimination. The major discriminatory metabolites consisted primarily of triterpenoid saponins and flavonoid glycosides. Saponin-associated variation was characterized by differences in glycosylation patterns and aglycone types, with medicagenic acid-type saponins containing five to six sugar moieties and medicagenic acid/yunganogenin K-type saponins containing three to four sugar moieties contributing to cultivar separation. Flavonoid-associated discrimination was mainly attributed to differences in spinacetin- and patuletin-derived glycosides.

These findings demonstrate substantial cultivar-dependent phytochemical diversity in spinach and identify characteristic saponin and flavonoid glycoside profiles as major contributors to metabolic differentiation among cultivars. This study provides a metabolomic basis for the characterization and selection of spinach cultivars with distinct phytochemical compositions and may contribute to the development of spinach germplasm with enhanced functional and nutritional value.

Keywords Flavonoid glycosides, Spinach (*Spinacia oleracea* L.), Triterpenoid saponins, Untargeted metabolomics, UPLC-QTOF/MS

P-16

Seasonal metabolomic variation and anticancer potential of bioactive metabolites in *Paulownia tomentosa* fruits

Su-Yeon Lee[1,2†], In-Seo Heo[1,2†], Hyoung-Geun Kim[1], Mun-Ock Kim[1], Inhyeok Song[3], Jaeha Kang[3], Jaewon Yang[3], Eun-Bin Kwon[1], Myung-Ji Kang[1], Doo-Young Kim[1], Jung-Hee Kim[1], Sei-Ryang-Oh[1], Daesik Jeong[4*], Eun Kyoung Seo[2*], Hyung Won Ryu[1*]

[1]*Natural Product Research Center and Natural Product Central Bank, KRIBB, 30-Yeongudanji-ro, Ochang-eup, Cheongwon-gu, Cheongju-si, Chungcheongbuk-do, 28116, Korea*
[2]*College of Pharmacy, Graduate School of Pharmaceutical Sciences, Ewha Womans University, 52, Ewhyeodae-gil, Seodaemun-gu, Seoul, 03760, Korea*
[3]*Bio Dx Group, InSiliCox, 202-ho, 42 Bongeunsa-ro 30-gil, Gangnam-gu, Seoul 06136, Korea*
[4]*Faculty of Software Convergence, Sangmyung University, 20 Hongjimun 2-gil, Jongno-gu, Seoul 03016, Korea*
[†]*These authors have equally contributed as the first author*

Paulownia tomentosa is a fast-growing deciduous tree traditionally used in herbal medicine for inflammatory and infectious diseases. Previous phytochemical studies have reported that *P. tomentosa* fruits are rich in phenylethanoid glycosides and *C*-geranylated flavonoids with diverse biological activities. However, dynamic changes in the anticancer potential of metabolites during fruit development remain poorly understood. This study aimed to identify characteristic metabolites and determine the optimal harvest time using UPLC-QTOF-MS-based metabolomics, and cancer cell inhibition assays.

Multivariate analyses revealed distinct clustering according to growth stages. A total of 36 metabolites were tentatively identified, including phenylethanoids/phenylpropanoids glycosides and *C*-geranylated flavonoids. Five marker compounds were isolated and structurally characterized using UPLC-QTOF-MS and $^{1}H/^{13}C$ NMR analyses. Quantitative analysis revealed seasonal differences, with compound **4** predominantly detected in July, compounds **2** and **3** enriched in August, and compounds **1** and **5** highest in September.

Diplacone and compound **4** exhibited strong cytotoxic effects against A549 and HepG2 cells, whereas compound **2** showed greater cytotoxicity toward HepG2 cells. Network-based target analysis identified inflammation-related hub genes, including PTGS2, IL6, TNF, and ALOX5. Notably, compound **4** uniquely showed direct TNF binding and associations with NFKB1 and CASP8, suggesting involvement of NF-κB signaling and apoptosis-related pathways.

These findings provide metabolomic and functional insights into seasonal phytochemical variation in *P. tomentosa* fruits and suggest optimal harvest periods for bioactive *C*-geranylated flavonoids.

Keywords *C*-geranylated flavonoids; Metabolomics; *Paulownia tomentosa*; Seasonal variation; UPLC-QTOF-MS

P-17

Comparative metabolomic profiling of *Lysimachia mauritiana* Lam. across cultivation environments: Differentially abundant metabolites and environment-specific biomarkers

Su-Ah Lee[1,2], Hyoung-Geun Kim[1], Hangyeol Lee[1], Doo-Young Kim[1], Su-Yeon Lee[1,2], Seul Beom Yu[1], Soo Ho Yeon[3], Hyoung Tae Kim[4], Jin-Hyub Paik[5], Sei-Ryang Oh[1], Eun Kyoung Seo[2*], Hyung Won Ryu[1*]

[1]*Natural Product Research Center and Natural Product Central Bank, KRIBB, 30-Yeongudanji-ro, Ochang-eup, Cheongwon-gu, Cheongju-si, Chungbuk, 28116, Republic of Korea*
[2]*College of Pharmacy, Graduate School of Pharmaceutical Sciences, Ewha Womans University, 52, Ewhyeodae-gil, Seodaemun-gu, Seoul, 03760, Korea*
[3]*Kolmar BNH, 6-Hanbang expo-ro, Jecheon-si, Chungcheongbuk-do, 27116, Republic of Korea*
[4]*College of Ecology and Environmental Science, Kyungpook National University, 2559, Gyeongsang-daero, Sangju, Gyeongsangbuk-do, 37224, Korea*
[5]*International Biological Material Research Center, KRIBB, 125-Gwahak-ro, Yuseong-gu, Daejeon, 34141, Republic of Korea*

Lysimachia mauritiana Lam. is an underutilized edible coastal plant belonging to the family Primulaceae. Its young shoots are consumed as a seasonal vegetable in Korea, but information on its cultivation-dependent metabolite variation and functional properties remains limited. In this study, *L. mauritiana* cultivated under three different environments—open field, plant factory, and glass greenhouse—was investigated to evaluate changes in secondary metabolites and antioxidant potential. Comparative metabolomic profiling was performed using an LC-MS/MS system, and multivariate statistical analysis revealed distinct metabolite patterns among cultivation environments. Differentially abundant metabolites (DAMs) were identified, and key biomarkers representing each cultivation condition were selected. In parallel, antioxidant activities and phytochemical contents were evaluated using 70% ethanolic extracts and Diaion HP-20 fractions. DPPH and ABTS radical scavenging assays were conducted with trolox as a positive control, and TPC, TFC, and TSC were determined using gallic acid, quercetin, and escin as reference standards, respectively. The extracts and fractions showed cultivation- and fraction-dependent differences in antioxidant activity and phytochemical contents. These results suggest that cultivation environment influences both the metabolite composition and functional properties of *L. mauritiana.* This study provides a scientific basis for establishing optimal cultivation conditions and supports the potential use of *L. mauritiana* as a functional food material.

Keywords Cultivation environment; Metabolomics; Antioxidant activity

P-18

Identification of tissue-specific chemical markers and antioxidant activities in different plant parts of *Vitex rotundifolia* L. f. using UPLC-QTOF/MS-based metabolomics

Seul Beom Yu[1,2], Hyoung-Geun Kim[1],Hangyeol Lee[1], Doo-Young Kim[1], In Soo Suh[3], Sung Chun Kim[3],Sei-Ryang Oh[1], Dongho Lee[2*],Hyung Won Ryu[1*]

[1]*Natural Product Research Center, Korea Research Institute of Bioscience and Biotechnology KRIBB, 30-Yeongudanji-ro, Ochang-eup, Cheongwon-gu, Cheongju-si, Chungbuk, 28116, Republic of Korea*
[2]*Department of Plant Biotechnology, College of Life Sciences and Biotechnology, Korea University, 145-Anam-ro, Seongbuk-gu, Seoul, 02841, Republic of Korea*
[3]*Biodiversity Research Institute, Clean Bio Business Division, JejuTechnopark, 338, Sinryedong-ro, Namwon-eup, Seogwipo-si, Jeju-do, 63608, Republic of Korea*

Vitex rotundifolia L. f. is a coastal halophyte belonging to the family Verbenaceae and is widely distributed throughout Korea, including Jeju Island, as well as in China, Japan, and Southeast Asia. The fruits of this plant have traditionally been used as a medicinal resource and is known to possess various biological activities, including antioxidant, anti-inflammatory, and analgesic effects. Previous phytochemical studies have reported that *V. rotundifolia* contains diverse secondary metabolites, such as flavonoids, diterpenoids, and iridoids. In this study, we investigated the major metabolite markers and antioxidant activities of different plant parts of *V. rotundifolia*, including leaves, stems, and seeds, were investigated. Extracts from each part were analyzed using UPLC–QTOF/MS, and multivariate analyses, including PCA, PLS-DA, and OPLS-DA, were performed to evaluate metabolic differences and identify tissue-specific chemical markers. Distinct metabolite profiles were observed among the different plant parts, enabling the characterization of representative markers metabolites associated with each tissue. In addition, antioxidant activity was evaluated using DPPH and ABTS radical scavenging assays, and it correlations between radical scavenging activities and phytochemical composition, including total phenolic content (TPC), total flavonoid content (TFC), and total saponin content (TSC), were further analyzed. The results provide useful insights into the phytochemical diversity and bioactive potential of *V. rotundifolia*, supporting its potential application as a functional material and natural product-based pharmacological resource.

Keywords Metabolite profiling, Tissue-specific markers, Antioxidant activity

P-19

Relationship between Chemical Composition and Biological Activity of Supercritical Fluid Extracted Arthropod Biomass

Won Jun Lee*

Ecological Technology Research Team, National Institute of Ecology, Seocheon-gun 33657, Republic of Korea

Supercritical carbon dioxide (SC–CO_2) extraction is an environmentally friendly technique for recovering lipophilic bioactive compounds from natural resources. In this study, SC–CO_2 extraction was applied to five edible or medicinal arthropod species (Protaetia brevitarsis seulensis, Zophobas morio, Gryllus bimaculatus, Ptecticus tenebrifer, and Scolopendra subspinipes mutilans) under standardized conditions (400 bar, 50 °C, and 180 min). Extraction yields were determined, and the chemical composition of the most active extract was characterized by GC–MS. Among the tested species, S. subspinipes mutilans exhibited the strongest biological activities despite its relatively low extraction yield, indicating that biological activity was more closely associated with chemical composition than extraction efficiency. GC–MS analysis revealed that the SC–CO_2 extract was enriched with fatty acids, fatty acid esters, and glyceride derivatives. Comparative screening of the identified constituents further showed that 4-hydroxyphenylacetic acid (4-HPAA) was the most promising lead compound, as it suppressed DOPA oxidation and intracellular melanogenesis without detectable cytotoxicity under the tested conditions. Several additional constituents also displayed anti-tyrosinase activity, indicating that the anti-melanogenic effect of the biomass extract is associated with a chemically definable constituent profile rather than a nonspecific crude extract effect.

Keywords *Scolopendra subspinipes mutilans*, Supercritical CO_2 extraction, GC–MS profiling

P-20

Contribution of Minor Coumarins to the Anti-inflammatory Activity of *Cnidium monnieri* Assessed by Knock-out Fractionation

Sangmin Lee[1,2], Sun Yong Park[2], Yerim Joo[1,2], Eunbeen Shin[1,2] and Seon Beom Kim[1,2,3]*

[1]*Department of Food Science and Technology, College of Natural Resources and Life Science, Pusan National University, Miryang 50463, Republic of Korea*

[2]*Institute for Future Earth, Pusan National University, Busan 46241, Republic of Korea*

[3]*Food Tech Innovation Center, Life and Industry Convergence Research Institue, Pusan national University, Miryang, Republic of Korea*

Atopic dermatitis (AD) is a chronic inflammatory skin disorder characterized by dysregulated inflammatory responses in the epidermis. *Cnidium monnieri* contains diverse bioactive coumarins, among which osthole and imperatorin are the major constituents. However, the contribution of less abundant coumarins to its anti-inflammatory activity remains inadequately characterized. Quantum mechanics-based quantitative NMR (QM-qNMR) quantified osthole and imperatorin at 16.42% and 6.90% (w/w), respectively, in the crude extract. Centrifugal partition chromatography (CPC) was subsequently employed for selective knock-out fractionation of osthole and imperatorin, generating compositionally defined samples for bioactivity assessment. Anti-inflammatory efficacy was evaluated in TNF-α-/IFN-γ-stimulated HaCaT keratinocytes as an in vitro AD model. The osthole–imperatorin knock-out fraction exhibited the most potent inhibitory activity, indicating that minor coumarins substantially contribute to the anti-AD bioactivity of *C. monnieri*, potentially via synergistic phytochemical interactions.

Keywords *Cnidium monnieri*, centrifugal partition chromatography, quantum mechanics-based quantitative NMR, anti-inflammatory

P-21

Chemical Characterization of Bioactive Sulfated Polysaccharides Isolated from *Stichopus japonicus*

Eunbeen Shin[1,2], Ibukunoluwa Fola Olawuyi[2], Sangmin Lee[1,2], Minhyuk Jeong[1,2], and Seon Beom Kim[1,2,3,*]

[1]*Department of Food Science and Technology, College of Natural Resources and Life Science, Pusan National University, Miryang 50463, South Korea*

[2]*Institute for Future Earth, Pusan National University, Busan 46241, South Korea*

[3]*Food Tech Innovation Center, Life and Industry Convergence Research Institute, Pusan National University, Miryang, South Korea*

Marine organisms represent a valuable source of structurally diverse natural products with unique chemical features and biological properties. Among them, sulfated polysaccharides (SPs) are complex marine-derived macromolecules characterized by diverse monosaccharide compositions, glycosidic linkages, and sulfation patterns. In this study, SPs were isolated and chemically characterized from the sea cucumber *Stichopus japonicus* to investigate their structural characteristics and potential as bioactive natural products. Raw samples were obtained through enzymatic digestion, selective precipitation, and ethanol precipitation, followed by purification using anion-exchange chromatography. Two major sulfated polysaccharide fractions (Fr. A and Fr. B) were successfully isolated, and their sulfated characteristics were confirmed by dimethylmethylene blue (DMMB) assay. Structural characterization was performed using 1D, 2D NMR spectroscopy, including 1H NMR, COSY, TOCSY, HSQC, HSQC-TOCSY, and NOESY experiments. Detailed NMR signal assignments identified Fr. A as fucosylated chondroitin sulfate (FucCS), mainly composed of fucose, N-acetylgalactosamine, and glucuronic acid residues, whereas Fr. B was characterized as a sulfated fucan (SF) predominantly consisting of sulfated fucose residues. Molecular weight distributions of both fractions were further determined by gel permeation chromatography. The purified FucCS and SF will be further evaluated for their anti-inflammatory activities. These findings provide detailed chemical information on structurally distinct SPs from *S. japonicus* and contribute to the characterization of marine sulfated polysaccharides as potential anti-inflammatory natural products.

Keywords Sulfated polysaccharides, *Stichopus japonicus*, Extraction and Purification, Characterization

P-22

Rapid Screening of Binding Constituents in Blueberry Extract Using Tyrosinase

Dayeon Kang[1], Hyoung-Geun Kim[2], and Hyung Won Ryu[2*]

[1]*Osong High School, Cheongju, Chungbuk, 28165, Republic of Korea*
[2]*Natural Product Research Center, Korea Research Institute of Bioscience and Biotechnology KRIBB, 30-Yeongudanji-ro, Ochang-eup, Cheongwon-gu, Cheongju-si, Chungbuk, 28116, Republic of Korea*

Tyrosinase는 멜라닌 생합성의 속도 결정 단계를 촉매하는 구리 함유 효소이다. 이 효소가 과도하게 활성화되면 피부 과색소침착이 일어나고, 과일과 채소에서는 효소적 갈변이 발생한다. 이 때문에 tyrosinase 저해제는 화장품, 의약품 및 식품 산업 전반에서 큰 관심을 받고 있다. 그러나 kojic acid, arbutin 등 널리 사용되는 저해제는 안정성과 안전성 측면에서 한계가 있어, 천연물에서 새로운 저해제를 찾으려는 수요가 계속되고 있다. 블루베리는 폴리페놀 화합물이 풍부하여 유망한 후보 소재이지만, 복잡한 식물 추출물에서 활성 성분을 찾아내기는 여전히 어렵다. 기존의 활성 유도 분획법은 분리와 활성 평가를 반복해야 하므로 시간과 노력이 많이 소요되며, 미량 활성 성분을 놓치기 쉽다. 이러한 한계를 극복하기 위해, 표적 단백질과 리간드 간의 특이적 상호작용을 이용하는 친화성 기반 스크리닝 기법이 조추출물에서 생리활성 성분을 직접 선별하는 효율적인 도구로 주목받고 있다. 본 연구에서는 친화성 ultrafiltration와 UPLC-PDA 분석을 결합하여 블루베리 추출물 내 tyrosinase 결합 성분을 신속하게 탐색하고자 하였다. 추출물을 버섯 유래 tyrosinase와 반응시킨 후 한외여과를 통해 비결합 분획을 분리하고 UPLC-PDA로 분석하였다. 효소 양에 따라 결합 조건을 최적화하였으며, 효소 처리군과 미처리군의 크로마토그램 피크 면적을 비교하여 결합 후보 성분을 선별하였다. Tyrosinase 처리 시 피크 면적이 감소한 6개 피크를 티로시나아제 결합 후보 성분으로 선별하였다. 이들 후보 성분은 UPLC-QTOF/MS로 추가 분석하였으며, 정확 질량, 예측 분자식 및 MS/MS 단편화 패턴을 문헌 보고 자료 및 온라인 데이터베이스와 비교하여 구조를 tentative identification 하였다. 본 접근법은 천연물 추출물에서 tyrosinase 표적 화합물을 신속하게 발굴하는 간편하고 효율적인 전략을 제시한다.

Keywords blueberry, tyrosinase, affinity ultrafiltration, UPLC-PDA, ligand screening, UPLC-QTOF/MS

P-23

Identification of Metabolites and Gene Expression Changes in *Vigna angularis* L. Sprouts under Indoor and Outdoor Cultivation Conditions Using HPLC-QTOF-MS and RT-PCR

Bo Gun Jeong[1], Han gyeol Lee[2], and Hyung Won Ryu[2*]

[1]*Cheongwon High School, 9 Guryong 5-gil, Ochang-eup, Cheongwon-gu, Cheongju 28121, Republic of Korea*
[2]*Natural Product Research Center and Natural Product Central Bank, KRIBB, 30-Yeongudanji-ro, Ochang-eup, Cheongwon-gu, Cheongju-si, Chungcheongbuk-do 28116, Republic of Korea*

주변 환경은 식물의 생장과 대사에 중대한 영향을 미친다. 본 연구에서는 재배 환경에 따른 팥순의 생육 특성 변화를 확인하기 위해 팥(*Vigna angularis* L.)을 실내와 실외에서 재배하여 팥순에 대한 표현형, 대사체 변화, 생장에 관한 유전자 발현 차이를 분석하였다. 재배 후 형태적 특성을 비교한 결과, 실외 재배 팥순은 실내 재배 팥순에 비해 줄기의 신장이 저조하였으나 줄기가 굵고 잎의 색이 진하였다. 이러한 형태적 차이와 함께 대사산물 조성의 변화를 확인하기 위해 HPLC-QTOF-MS 분석을 통해 실내 재배 팥순과 실외 재배 팥순의 대사산물을 분석하였고, flavonoid계열의 kaempferol 배당체 5종과 Coutaric acid를 확인하였다. 이 화합물들의 피크 면적값을 비교한 결과 flavonoid 계열 화합물의 피크 면적값이 실외 재배 팥순에서 실내 재배 팥순보다 2배 이상 높은 면적을 나타냈다. 또한 이러한 재배 환경에 따른 생육 특성의 차이를 분자 수준에서 확인하기 위해 성장과 관련된 Auxin-responsive protein IAA17(*IAA17)* 유전자와 강한 빛에 반응하는 Light-Harvesting complex II chlorophyll a/b binding protein 1(*LHCB1*) 유전자를 RT-PCR로 분석하였다. *IAA17* 유전자는 줄기 신장이 활발한 실내 재배 팥순에서 높게 발현 수준을 보였으며, *LHCB1* 유전자는 높은 광량에 노출된 실외 팥순에서 높게 발현되였다. 이상의 결과는 낮은 광량의 재배 환경에서는 팥순의 줄기 신장을 촉진하는 반면 광량이 높은 환경은 줄기의 비대와 Flavonoid의 상대적 축적을 증가시키며, 재배 환경에 따라 생장 및 광 반응 관련 유전자의 발현 양상이 달라질 수 있음을 보여준다. 본 연구에서는 재배 환경이 팥순의 생육 특성과 대사산물 조성에 영향을 미치는 주요 요인임을 확인하였다.

Keywords *Vigna angularis* L., HPLC-QTOF-MS, RT-PCR, Cultivation conditions

P-24

A Study on Functional Materialization Using Persimmon-Peel Biomass and the Development of Cosmetics: Optimization of Enzymatic Fermentation and Ultra-Fine Milling Extraction Processes and Efficacy Evaluation

Yu-Rim Lee, Mi-Ae Do, Hwa-Jin Suh*

Convergence R&D Center, Korea Textile Smart Process Institute, Gongju 31156, Republic of Korea

Persimmon peel, which is generated in large quantities during persimmon processing, is mostly discarded, aggravating resource waste and increasing the burden of waste treatment. This study is a preliminary investigation examining the conditions for materializing discarded persimmon peel as a raw material for bio-additives used in the manufacture of synthetic leather. Persimmon peel — which contains large amounts of functional constituents such as tannins and polyphenols — was subjected to enzymatic fermentation and ultra-fine milling extraction, optimal extraction conditions were established, and the functionality of the extract was evaluated. Fermivin was applied to a persimmon homogenate for fermentation at 40 °C for 2 hours, and Sumizyme AC enzyme was applied for a fermentation reaction at 55 °C for 2 hours. Eight extraction conditions were then compared, combining distilled water and 50% ethanol as solvents with conventional blender extraction and nano-scale ultra-fine milling extraction. Under all conditions, the ultra-fine milling extraction showed higher recovery yields and total polyphenol contents than blender extraction. In addition, the edible portion showed a higher recovery yield than the non-edible portion; however, in this study, the ultra-fine extraction of the non-edible portion with 50% ethanol was selected as the optimal condition (total polyphenol 64.7 GAE mg/g). The extract obtained under the optimal condition exhibited antioxidant activity approaching that of vitamin C; DPPH radical scavenging of 81.8% (16 μg/mL) and xanthine oxidase inhibition of 73.1% (160 μg/mL); and also exhibited wrinkle-improvement-related efficacy, with elastase inhibition of 91.3% (160 μg/mL). A cream-formulation prototype secured good stability over a 4-week harsh (accelerated) test. These results suggest that a low-temperature extraction process combining enzymatic fermentation and ultra-fine milling is an effective technology for the valorization of persimmon by-products into cosmetic materials.

Keywords Persimmon by-product, Enzymatic fermentation, Nano-milling extraction, Total polyphenol, Antioxidant, Cosmetic material

Acknowledgement 본 연구는 산업통상부(한국산업기술진흥원)의 저탄소기후변화대응을위한지속성장(R&D)사업의 일환으로 수행하였음 [과제고유번호 : RS-2026-25549673, 감 가공 부산물의 기능성 소재화를 통한 바이오매스 기반 친환경 인조피혁 제품화].

P-25

Anti-inflammatory Activity of Cheongyeolhaedok-tang via the Regulation of Inflammatory Chemokines and Antimicrobial Peptides in a Psoriasis-Like Human Keratinocyte Model

[1]Tae-hee Go, [1]Na-Yeon Kang, [2]Yeop Jung, [2]Sun-Young Park, [1]Hyung-Seo Hwang*

[1]Department of Bio-Cosmetic Science, Semyung University, Jecheon, Chungbuk 27136, South Korea
[2]Department of Physiology, College of Korean Medicine, Semyung University, Jecheon, Chungbuk 27136, South Korea

Psoriasis is a chronic inflammatory skin disease characterized by keratinocyte hyperproliferation and immune dysregulation. Cheongyeolhaedok-tang (淸熱解毒湯; CYHDT), a traditional herbal prescription documented in *Donguibogam* (東醫寶鑑), has long been used to treat intractable inflammatory skin diseases; however, its molecular mechanisms in chronic skin inflammatory pathogenesis, such as psoriasis, remain to be fully elucidated. Therefore, this study aimed to scientifically validate the anti-inflammatory and psoriasis-alleviating effects of CYHDT using an *in vitro* cell model. CYHDT was prepared via hot water extraction of medicinal herbs, including *Scutellaria baicalensis*, *Coptis chinensis*, and *Lonicera japonica*, according to the prescription in *Donguibogam*. To evaluate the antioxidant activity of CYHDT, total polyphenol content (TPC) and DPPH/ABTS radical scavenging assays were performed, demonstrating high radical scavenging and antioxidant capacities. In addition, evaluation of nitric oxide (NO) production in RAW 264.7 cells revealed that CYHDT treatment significantly suppressed NO generation. Subsequently, human keratinocytes (HaCaT) were stimulated with TNF-α, IFN-γ, IL-17A, and IL-22 to induce a psoriasis-like inflammatory condition, followed by treatment with CYHDT at various concentrations. RT-qPCR analysis revealed that CYHDT significantly suppressed the mRNA expression of chemokines (CXCL8, CCL20), a pro-inflammatory cytokine (IL-1β), and an antimicrobial peptide (hBD-2), all of which were upregulated under the psoriatic conditions. Overall, CYHDT effectively alleviated abnormal inflammatory responses and the antimicrobial peptide over-expression in keratinocytes. These findings provide a scientific foundation for the clinical application of traditional herbal prescription and highlight CYHDT as a promising natural bio-material for alleviating psoriatic skin inflammation.

P-26

Protective Effects of *Geranium thunbergii* Extract Against UV-Induced Skin Aging in Human Dermal Fibroblasts and Zebrafish

Ye-Jin Lee[1,3], Bongkyun Park[1,3], Heung Joo Yuk[2,3] and Young Sook Kim[1,3,*]

[1]*Korean Medicine Convergence Research Division, Korea Institute of Oriental Medicine, Daejeon 34054, Republic of Korea*
[2]*NeuroGlymph Imaging and Modulation Center, Korea Institute of Oriental Medicine, Daejeon 34054, Republic of Korea*
[3]*School of Korean Convergence Medical Science, University of Science and Technology (UST), Daejeon, 34113, Republic of Korea*

Ultraviolet (UV) radiation is a major extrinsic factor that contributes to skin aging by inducing oxidative stress, cellular senescence, inflammation, and extracellular matrix degradation. This study investigated the protective effects of *Geranium thunbergii* (GT) extract against UV-induced skin aging using human dermal fibroblasts (HDFs) and a zebrafish model. Chemical profiling of GT extract by UPLC-PDA and ESI-qTOF-MS identified corilagin, ellagic acid, and kaempferitrin as major constituents. In UV-exposed HDFs, the expression of senescence-related genes, including p16, p21, and p53, was increased, whereas GT treatment at 80 μg/mL attenuated these UV-induced changes. GT treatment also reduced SA-β-gal-positive area and intracellular reactive oxygen species (ROS) levels by approximately 73.0% and 79.7%, respectively, compared with the UV-treated group. In UVB-exposed zebrafish, GT treatment reduced ROS production by approximately 35.6% and 40.4% at 50 and 100 μg/mL, respectively, compared with the UVB-treated group. Consistently, qualitative SA-β-gal staining showed an apparent reduction in senescence-associated β-galactosidase activity, with the most pronounced effect observed at 100 μg/mL. At the molecular level, UVB exposure altered the expression of senescence and photoaging related genes, including p53, MMP2, MMP9, and IL-1β. GT treatment showed a reducing effect on the expression of these genes, particularly at 100 μg/mL. In summary, these findings suggest that *G. thunbergii* extract exerts protective effects against UV-induced skin aging by suppressing oxidative stress and cellular senescence and by modulating photoaging-related molecular responses in both HDFs and zebrafish. These results support the potential of GT as a natural candidate for the prevention or improvement of UV-induced skin aging.

Keywords *Geranium thunbergii*, Photoaging, Cellular senescence

P-27

Screening of Psoriasis-Related Candidate Materials from Vietnamese Medicinal Plants Based on Antioxidant Activity and Total Phenolic Content

Jae Jun Kim, Yeon Woong Jung, Hyung Seo Hwang*

Department of Bio-Cosmetic Science, Semyung University, Jecheon, Chungbuk 27136, South Korea

Psoriasis is a representative chronic inflammatory skin disease, and an imbalance between oxidative stress and the antioxidant defense system has been reported to contribute to its pathogenesis. In addition, reactive oxygen species (ROS) generated by oxidative stress may be associated with inflammatory signaling pathways, including NF-κB, MAPK, and STAT3, thereby contributing to psoriatic inflammatory responses. Accordingly, this study aimed to identify novel materials for psoriasis by establishing antioxidant activity as a primary screening criterion and conducting a stepwise screening process. First, the antioxidant activities of 70 plant species traditionally used for medicinal purposes in Vietnam were evaluated using a DPPH radical scavenging assay, and 10 candidates exhibiting high antioxidant activity were selected as the primary candidate group. Subsequently, the antioxidant activities of the selected 10 candidates were reevaluated using an ABTS radical scavenging assay, and the consistency of the results with those obtained from the DPPH radical scavenging assay was assessed. Finally, based on the results of total phenolic content (TPC) analysis and antioxidant activity, *Mezoneuron cucullatum* (Vi 41-25) was selected as the final candidate material. Further studies are currently underway using a psoriasis-like cellular model to evaluate the biological activity of *Mezoneuron cucullatum* (Vi 41-25) and its potential effects on psoriasis.

P-28

Isolation and Structural Characterization of Khellactone-Type Coumarins from *Peucedanum japonicum* with Potential Anti-Neuroinflammatory Activity

Jeong Hyun Park[1], Woo-Cheol Shin[1], Jin-Kyu Jang[1], Hyeon Seon Na[1], Yong Jae Jeong[1], Dong-Sung Lee[2] and Dae Young Lee[1*]

[1]*BK21 FOUR KNU Creative BioResearch Group, School of Life Sciences, Kyungpook National University, Daegu 41566, Republic of Korea*
[2]*Research Institute of Pharmaceutical Sciences, College of Pharmacy, Chosun University, Gwangju 61452, Republic of Korea*

This study combined phytochemical characterization, cell-based evaluation, and computational target analysis to investigate the khellactone constituents of *Peucedanum japonicum* aerial parts and their potential anti-neuroinflammatory activity. Chromatographic separation of the ethyl acetate fraction yielded eighteen khellactone-type dihydropyranocoumarins (1–18). Their structures were established through combined 1D/2D NMR, high-resolution UPLC-QTOF-MS^E fragmentation, and electronic circular dichroism (ECD) analysis, which confirmed a common (3′S,4′S) absolute configuration. In cell-based assays, several isolates inhibited lipopolysaccharide-induced nitric oxide production in BV2 microglial cells, with IC_{50} values of 5.5–6.8 μM, whereas no clear protective effects were observed in HT22 cells.

Candidate molecular targets were further explored using network pharmacology and molecular docking. Among the 72 predicted targets common to the compounds, 50 overlapped with neuroinflammation-associated targets. Subsequent protein-protein interaction analysis identified STAT3, BCL2, HIF1A, GSK3B, PPARA, and MMP9 as six hub genes, with molecular docking predicting a favorable interaction between compound 13 and PPARA. Together, these findings characterize the khellactone constituents of *P. japonicum* and identify isolates with potential anti-neuroinflammatory activity, while highlighting candidate molecular targets.

Keywords *Peucedanum japonicum,* Khellactones, Network pharmacology, Molecular docking, Anti-neuroinflammatory

P-29

Regional Metabolomic Differentiation of *Perilla frutescens* Leaves Based on LC-Q-TOF/MS and Multivariate Analysis

Na Rae Kang[a,b], Yun Gon Son[a,b], Seungjae Jang[a,b], Seungyu Lee[a,b] and Jeong Yoon Kim[a,b,*]

[a]*Department of Pharmaceutical Engineering, ABC-RLRC, IALS, Gyeongsang National University, Jinju, 52725, Republic of Korea*

[b]*Gyeongnam Aerospace & Defense Institute of Science and Technology, Gyeongsang National University, Jinju 52828, Republic of Korea*

Perilla frutescens Britton var. *acuta* Kudo leaves are widely consumed in East Asia due to their culinary and medicinal properties, which are largely attributed to their high levels of bioactive metabolites such as rosmarinic acid. In this study, we investigated the variation in rosmarinic acid content and overall metabolite profile of *P. frutescens* leaves collected from six different provinces in Republic of Korea. Quantitative analysis of the rosmarinic acid content was performed using HPLC, revealing significant regional differences, with the highest concentration observed in the leaves collected in Gyeongsangbuk-do and the lowest concentration in those from Jeollanam-do. HRESIMS and 1H-NMR spectrometry were used to determine the chemical structure of the isolated rosmarinic acid. LC-Q-TOF/MS analysis identified ten major metabolites, including phenolic acids, flavonoids, and triterpenoids. Multivariate statistics (OPLS-DA) revealed distinct clustering of populations, indicating a strong relationship between metabolites and environmental parameters. The distribution of the metabolite fingerprints and rosmarinic acid contents in *P. frutescens* leaves were also found to differ according to the cultivation region, suggesting that secondary metabolite expression is influenced by environmental and geographic factors. This work shows that metabolome profiles can be used in quality control and the development of high-quality products derived from *P. frutescens*.

P-30

Tyrosinase Inhibitory Activity of Compounds Isolated from *Agastache rugosa* Seeds as Potential Natural Skin-Whitening Agents

Seungjae Jang[a], Seungyu Lee[a], Hoil Kim[a], Jeong Yoon Kim[a,*]

[a]*Department of Pharmaceutical Engineering, ABC-RLRC, IALS, Gyeongsang National University, Jinju, 52725, Republic of Korea*

Agastache rugosa, a member of the Lamiaceae family, is widely used as a food, spice, and medicinal resource in East Asia, including Korea, and has been reported to contain various flavonoids and phenolic compounds, such as acacetin and rosmarinic acid. This study aimed to evaluate the tyrosinase inhibitory activity of compounds isolated from the seeds of *Agastache rugosa* and to investigate their potential as natural skin-whitening agents. Two compounds were purified from the methanolic crude extract of *A. rugosa* seeds using silica gel and Sephadex chromatography. All isolated compounds were structurally elucidated based on spectroscopic data, including NMR, UV, and LC-Q-TOF/MS analyses. In the tyrosinase inhibitory assay, compound 1 and compound 2 exhibited IC_{50} values of 45.44 μM and 59.06 μM, respectively. Both compounds showed competitive inhibition against tyrosinase. These results suggest the potential of *A. rugosa* seeds as a natural source of skin-whitening agents.

P-31

Anti-biofilm Potential of Furanocoumarins Isolated from *Peucedanum japonicum* Seeds against *Pseudomonas aeruginosa*

Jihyeon Choi[a], Hoil Kim[a], Jeong Yoon Kim[a,*]

[a]*Department of Pharmaceutical Engineering, ABC-RLRC, IALS, Gyeongsang National University, Jinju, 52725, Republic of Korea*

Pseudomonas aeruginosa is an opportunistic pathogen capable of forming biofilms, which protect bacterial cells from antimicrobial agents and host immune responses, thereby increasing antimicrobial resistance and complicating the treatment of infections. The emergence of multidrug-resistant strains has led to increasing efforts to identify novel biofilm-control agents from natural products and their derivatives. *Peucedanum japonicum* is an edible and medicinal plant rich in coumarin and furanocoumarin derivatives; however, the anti-biofilm activities of individual compounds isolated from its seeds have not been sufficiently investigated. In this study, oxypeucedanin (1) and isoimperatorin (2) were isolated from *P. japonicum* seeds, and their structures were identified using HRESIMS and NMR analyses. Fluorescence microscopy of *P. aeruginosa* biofilms treated with the two compounds revealed reduced bacterial aggregation and alterations in biofilm architecture. In addition, molecular docking analysis targeting LasR, a quorum-sensing regulatory protein, was performed to investigate potential molecular interactions with the quorum-sensing system. These findings demonstrate the anti-biofilm potential of oxypeucedanin and isoimperatorin isolated from *P. japonicum* seeds and support their further investigation as natural-product-based candidates for controlling *P. aeruginosa* biofilms.

P-32

Tyrosinase Inhibition and Kinetic Characterization of Tracheloside and Arctiin from *Synurus deltoides* Seeds

Bashir Shanza[a], Seungmin Oh[a], Juhyeon Kim[a], Jeong Yoon Kim[a,*]

[a]*Department of Pharmaceutical Engineering, ABC-RLRC, IALS, Gyeongsang National University, Jinju, 52725, Republic of Korea*

Synurus deltoides, a perennial herb belonging to the Asteraceae family, has traditionally been used as an edible and medicinal resource. However, studies on the bioactivity and metabolites of its seeds remain limited. In this study, major compounds were isolated from *S. deltoides* seeds, and their tyrosinase inhibitory activity and skin-whitening effects were evaluated. The methanolic extract was suspended in water and successively partitioned with n-hexane and ethyl acetate. The major compounds were isolated and purified through MPLC. The isolated compounds were identified as tracheloside (SD1) and arctiin (SD2) by HPLC and LC-Q-TOF/MS analyses. In the tyrosinase inhibitory assay, tracheloside (SD1) and arctiin (SD2) exhibited IC_{50} values of 18.36 μM and 12.88 μM, respectively. Enzyme kinetic analysis revealed that tracheloside showed competitive inhibition, whereas arctiin exhibited mixed-type inhibition. These findings suggest that tracheloside and arctiin derived from *S. deltoides* seeds possess potent tyrosinase inhibitory activity and may serve as promising natural skin-whitening agents and functional cosmetic ingredients.

MEMO

MEMO

MEMO

MEMO

MEMO



한국응용생명화학회
The Korean Society for Applied Biological Chemistry